Electronic and Telecommunication Engineering

Section 4

MRI: Magnetic Resonance Imaging

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BM3122- Medical Imaging
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Introduction

- No ionizing radiation involved
- Cross-sectional images of anatomy or functionality
- Uses differences in magnetic properties of atomic nuclei
  - Provides a spatial map of the hydrogen nuclei (water and lipid) in different tissues.

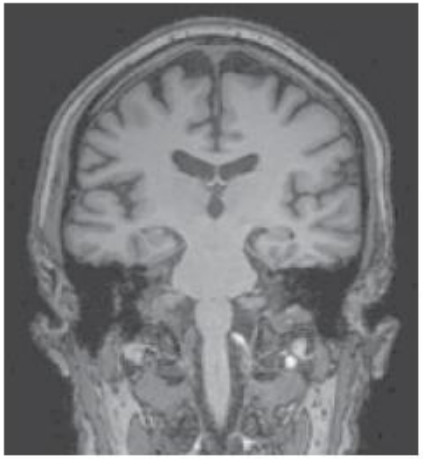
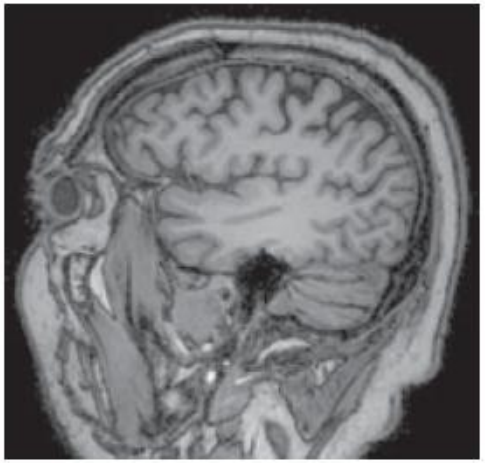
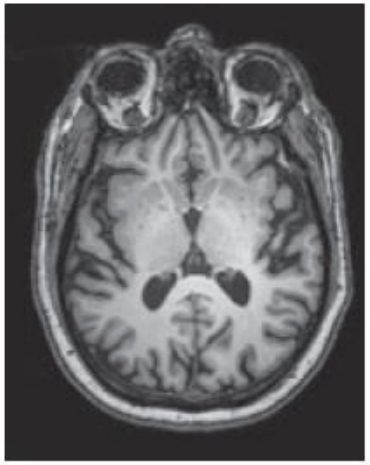
Introduction cont.

- The image intensity depends on
  - The number of protons in any spatial location
  - Physical properties of the tissue such as viscosity, stiffness and protein content
  - Data acquisition method

- MR images can be made sensitive to
  - Blood flow (angiography) and blood perfusion
  - Water diffusion
  - Localized functional brain activation (fMRI)

Introduction cont.

- Advantages of MRI are:
  1. No ionizing radiation is required
  2. Images can be acquired in any two- or three-dimensional plane
  3. Excellent soft-tissue contrast
  4. A spatial resolution less than 1 mm
  5. Negligible penetration effects

Introduction cont.

- The main disadvantages of MRI
  - MR image acquisition is much slower than other modalities
  - Some patients are precluded from MRI scans due to metallic implants from previous surgeries
  - Systems are much more expensive

Applications

- Pathologies in all parts of the body can be diagnosed:
  - Central Nervous System (CNS)
  - Internal organs
  - Vascular system
  - Heart
  - Musculoskeletal system
  - Lungs
  - …

Magnetic Properties of Nuclei

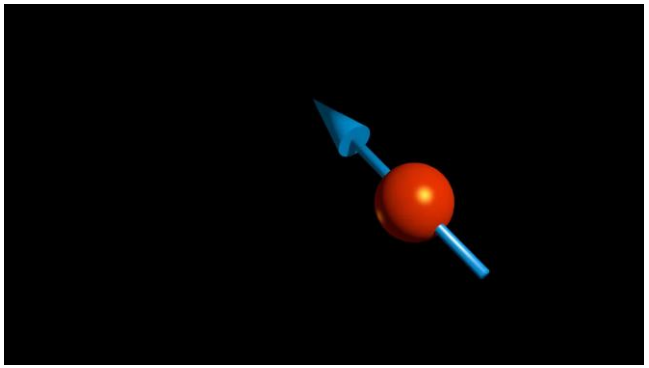
- Nuclear magnetic resonance (NMR) first observed by Bloch and Purcell in 1946
  - Atomic nuclei consist of protons and neutrons (nucleons)
  - All nucleons display a natural rotation
  - If even number of protons and neutrons
    - Nucleons are arranged in the opposite directions
    - Their spins cancel off
  - If nucleons are unpaired (odd number)
    - The nuclear spin is greater than zero
    - Spinning nuclei are referred to as “spins”

Magnetic Properties of Nuclei cont.

- Since nuclei are charged positively
  - The spinning acts as a circular current
  - Induces a weak magnetic field
    - Nuclei behave like small bar magnets
    - They possess a magnetic dipole moment
- Eg. ^{1}H, ^{13}C, ^{31}P have a dipole moment
  - In biological system the most abundant spin is ^{1}H (proton) in;
    - Water
    - Lipid

Magnetic Dipoles in a Magnetic Field

- Spins are in a random distribution
  - Pointing in all spatial directions
  - Net magnetic moment is zero
- When placed in a static magnetic field (B_0)
  - Magnetic moment vectors (spins) align with direction of B_0
    - In parallel or anti-parallel
  - Alignment generates a net magnetic moment **μ**
    - The net magnetization pointing in the direction of B_0
    - Angular momentum (**P**)

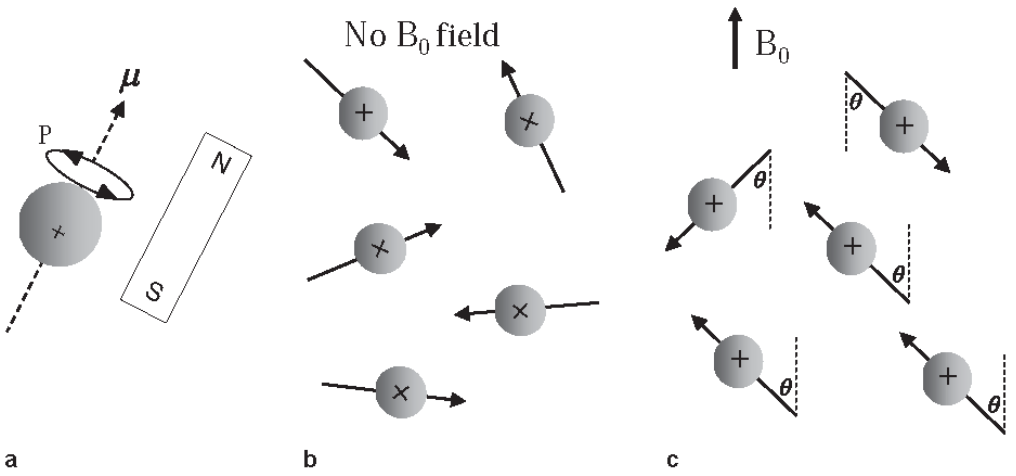


Magnetic Dipoles in a Magnetic Field cont.

(a) The internal rotation of a proton creates a magnetic moment

(b) Magnetic moments are completely random

(c) In the presence of a strong magnetic field magnetic moments align at an angle, $\theta=54.7^{o}$ the direction of B_0



Proton Energy Levels

- Magnitude of the angular momentum of the proton is quantized and has a single, fixed value
- Magnitude of the proton's magnetic moment is proportional to the magnitude of the angular momentum:

$$\left|\vec{\mu}\right| = \gamma\left|\vec{\mathrm{P}}\right|$$

Where γ = Gyro-magnetic ratio (depends on the nucleus involved, in *rad* s^{-1} T^{-1})

For protons, $\gamma = 267.52 \times 10^6$ rad s^{-1} T^{-1}

Proton Energy Levels cont.

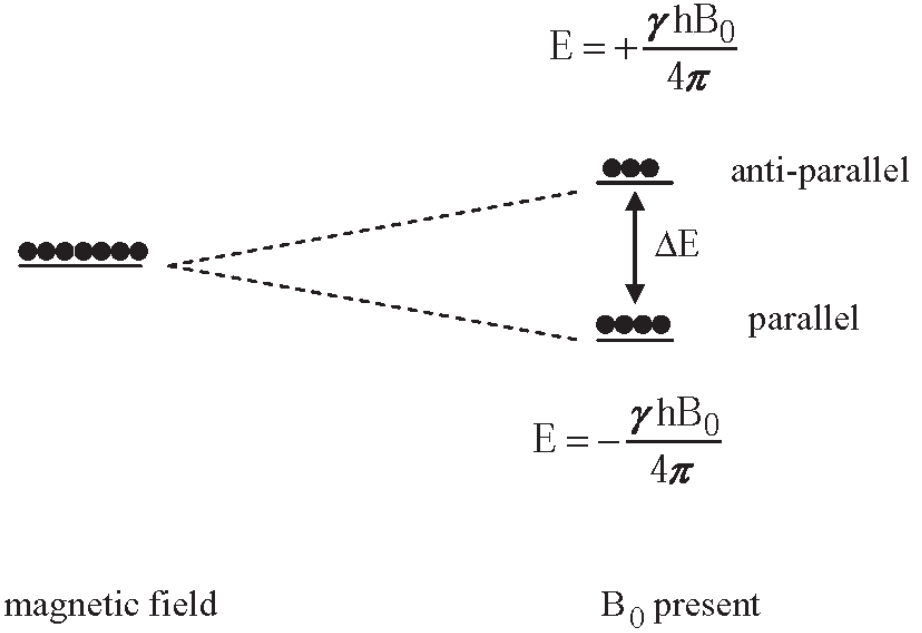
- The relative number of protons in the parallel and anti-parallel configurations
  - Depends on the value of B_0
  - Protons in the parallel configuration are in the lower energy state
- The energy difference between the two states:

$$\Delta E = \frac{\gamma h B_0}{2\pi}$$

  - Where *h* (Plank's constant) = 6.63 X 10^{-34} Js

Proton Energy Levels cont.

- Proton configurations
  - When a strong magnetic field is applied, the single energy level splits into two levels
  - There are slightly more protons in the parallel ground energy state than in the higher energy anti-parallel state



Proton Energy Levels cont.

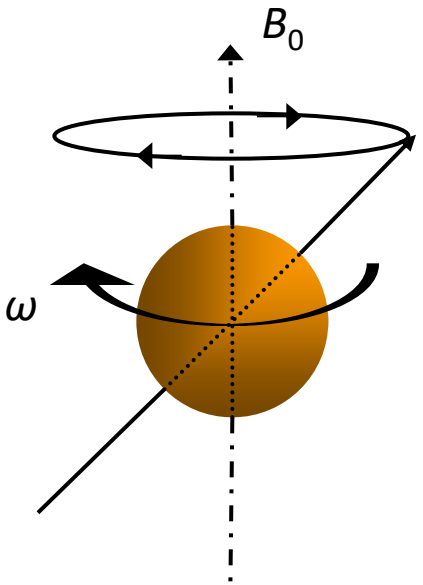
- The MRI signal depends upon the difference in populations between the two energy levels:

$$N_{parallel} - N_{anti\text{-}parallel} = N_{total} \frac{\gamma h B_0}{4\pi kT}$$

  - Where *k* is Boltzmann's constant with a value of 1.38×10^{-23} J/K and *T* is the temperature measured in Kelvin

Larmor Precession

- In the presence of a magnetic field, B_0
  - Spins are tilted with respect to B_0
  - Rotate around the axis of B_0
  - This motion is called Larmor Precession
  - Occurs at a precise frequency, determined by:
    - The magnetic field strength, B_0 (in *T*)
    - The characteristics of the nucleus



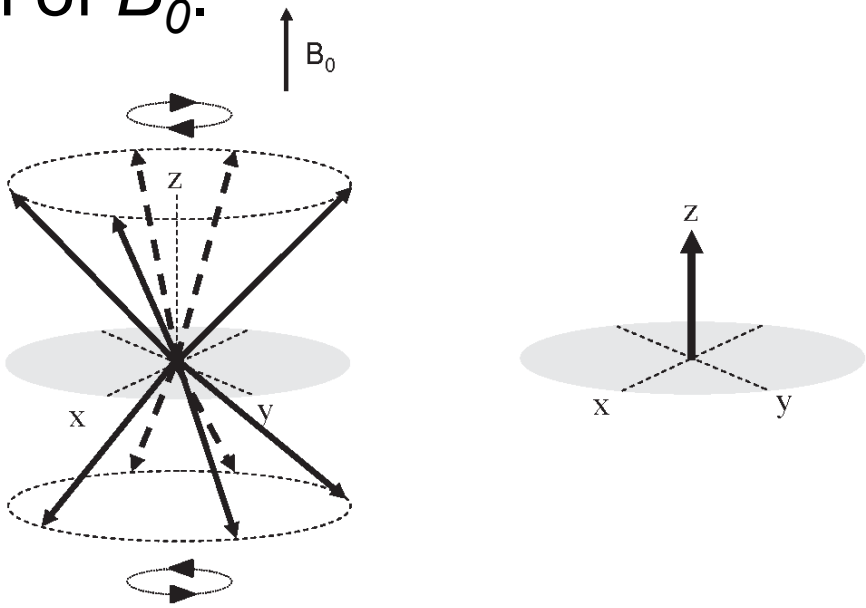
$$\omega_0 = \gamma B_0$$

ω_0 = Larmor frequency (in *rad/s*)

1*T* = 10^4 Gauss (Earth's magnetic field is 0.5 Gauss)

Larmor Precession cont.

- Magnetization represented by vectors
  - Individual magnetization vectors are randomly distributed around a cone which subtends an angle of 54.7° with respect to the B_0 (z) axis.
  - The vector sum of all vectors creates a static component in the direction of B_0.



Larmor Precession cont.

- The net magnetization, $\mathbf{M}_0$
  - Has only a z-component, since the vector sum of the x and y axes components is zero
  - Defined as:

$$M_0 = \sum_{n=1}^{N_{total}} \mu_{z,n} = \frac{\gamma h}{4\pi}\left(N_{parallel} - N_{anti\text{-}parallel}\right) = \frac{\gamma^2 h^2 B_0 N_{total}}{16\pi^2 kT}$$

Effects of a Radiofrequency Pulse

- The energy levels for protons in a magnetic field
  - Discrete
  - Analogous to energy levels in semiconductors
- To obtain an MR signal
  - Energy must be supplied with a specific value, (*ΔE)* to stimulate transitions between the energy levels
  - Frequency (f) of the energizing signal can be calculated from De Broglie's relationship, *ΔE=hf*;

$$\mathrm{hf} = \Delta\mathrm{E} = \frac{\gamma \mathrm{h} \mathrm{B}_0}{2\pi}$$

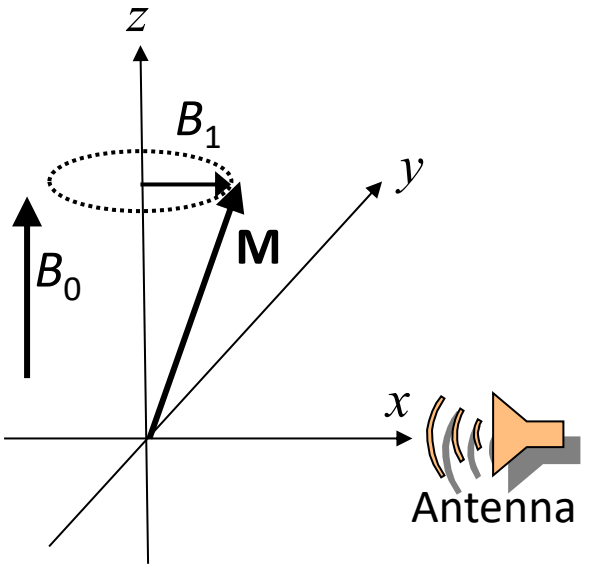
$$\Rightarrow \mathrm{f} = \frac{\gamma \mathrm{B}_0}{2\pi} \text{ or } \omega = \gamma \mathrm{B}_0$$

Effects of a Radiofrequency Pulse cont.

- The energy is supplied as an electromagnetic (EM) wave
  - Referred to as a radiofrequency (RF) field
  - Frequency of the RF field is identical to the precession (Larmor) frequency

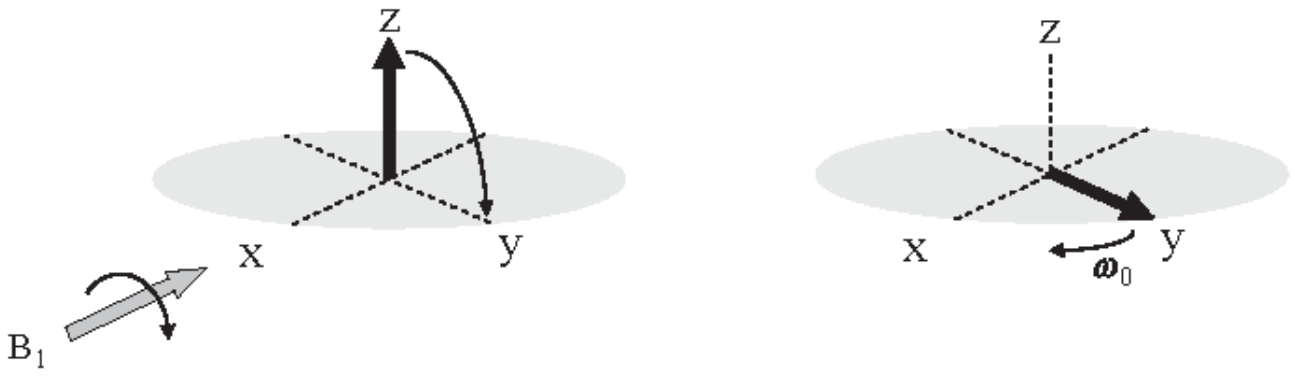
Creation of Transverse Magnetization

- In MRI, energy is applied as a short RF pulse
  - Direction of the magnetic component of the RF field oriented at 90° to the direction of B_0
  - Magnetic component of the RF pulse (B_1 field), produces a torque which causes the net magnetization to rotate towards the *xy*-plane



Creation of Transverse Magnetization cont.

- B_1 field
  - If applied along the *x*-axis, the magnetic moment vector is rotated towards the *y*-axis



Creation of Transverse Magnetization cont.

- The 'tip angle' (α)
  - Defined as the angle through which the net magnetization is rotated
  - Proportional to both the strength of the applied RF field (measured in Tesla) and the time, τ_{B1}

$$\alpha = \gamma \mathrm{B}_1 \tau_{\mathrm{B1}}$$

- A tip angle of 90° (termed a 90° pulse)
  - Results in the maximum value of the M_y component of magnetization
- A 180° pulse
  - No M_y magnetization, but rotates the net magnetization M_0 from the +z to the -z axis

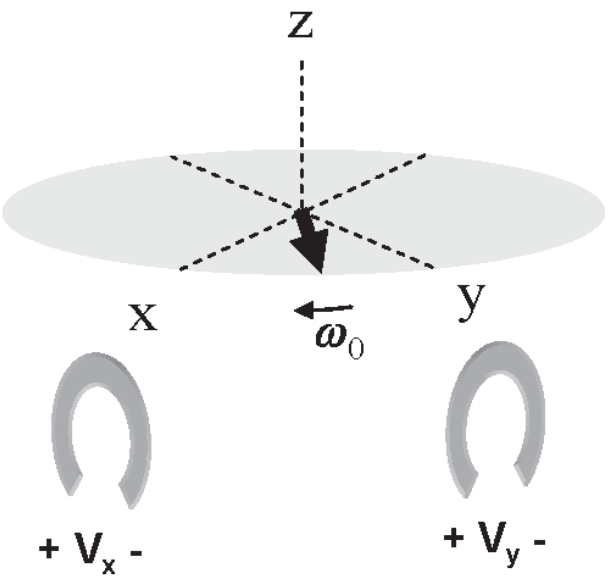
Faraday Induction

- Basis of MR signal detection
- Most simple case of the MR detector
  - Consists of a pair of conductive loops placed close to the object at 90° to each other
- Faraday's law of induction
  - Voltage (V) induced in loop is proportional to the time rate of change of the magnetic flux dφ:

$$V \propto -\frac{d\varphi}{dt}$$

Faraday Induction cont.

- The MR signal is measured via Faraday induction
  - Either one or two RF coils can be used
  - A voltage is induced by the precessing magnetization after a 90º pulse has been applied



Faraday Induction cont.

- Voltages induced in the two coils are given by:

$$V_y \propto M_0 \omega_0 \sin \omega_0 t$$

$$V_x \propto -M_0 \omega_0 \cos \omega_0 t$$

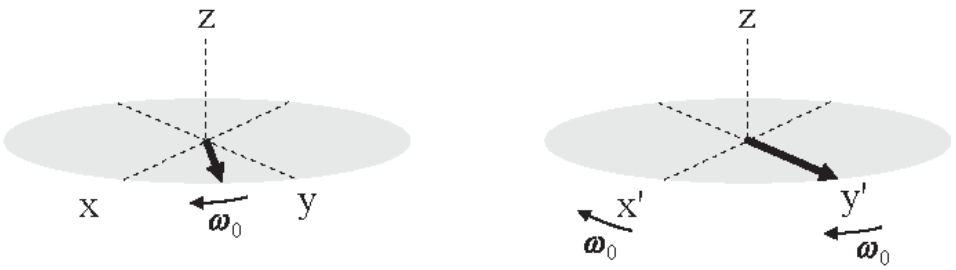
- For an induced MR signal (voltage)
  - A time-varying magnetic flux is required
  - Only magnetization precessing in the *xy*-plane is relevant
  - Any z-component of magnetization does not precess and therefore does not induce a voltage

MR Signal Intensity

- Determined by three different factors
  - Number of protons in the object (number of protons in each voxel of the image)
  - Value of M_0 (proportional to B_0)
  - Induced voltage (proportional to the precession frequency, which in turn is proportional to B_0)
- Overall, MR signal is proportional to the square of the B_0 field
  - Strong drive towards higher field MRI systems
  - Overall SNR depends on other factors too

The Rotating Reference Frame

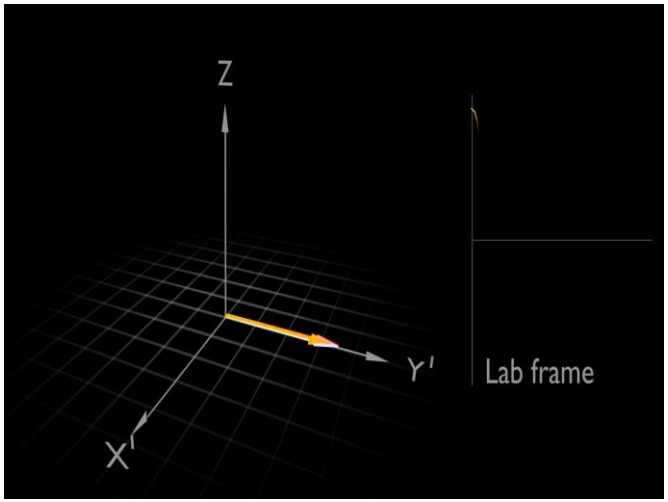
- In the stationary frame
  - Magnetization precesses in the transverse plane (M_x and M_y) around the z-axis at an angular frequency of ω_0 rad/s
- In the rotating reference frame
  - Axes rotate around the z-axis at ω_0 rad/s, and therefore the magnetization appears stationary
  - For easy visualization

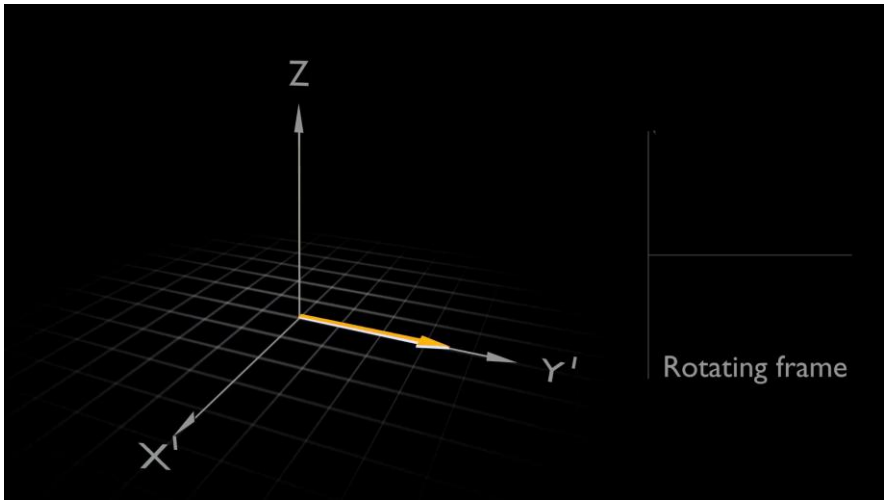


Stationary frame

Rotating reference frame

The Rotating Reference Frame cont.



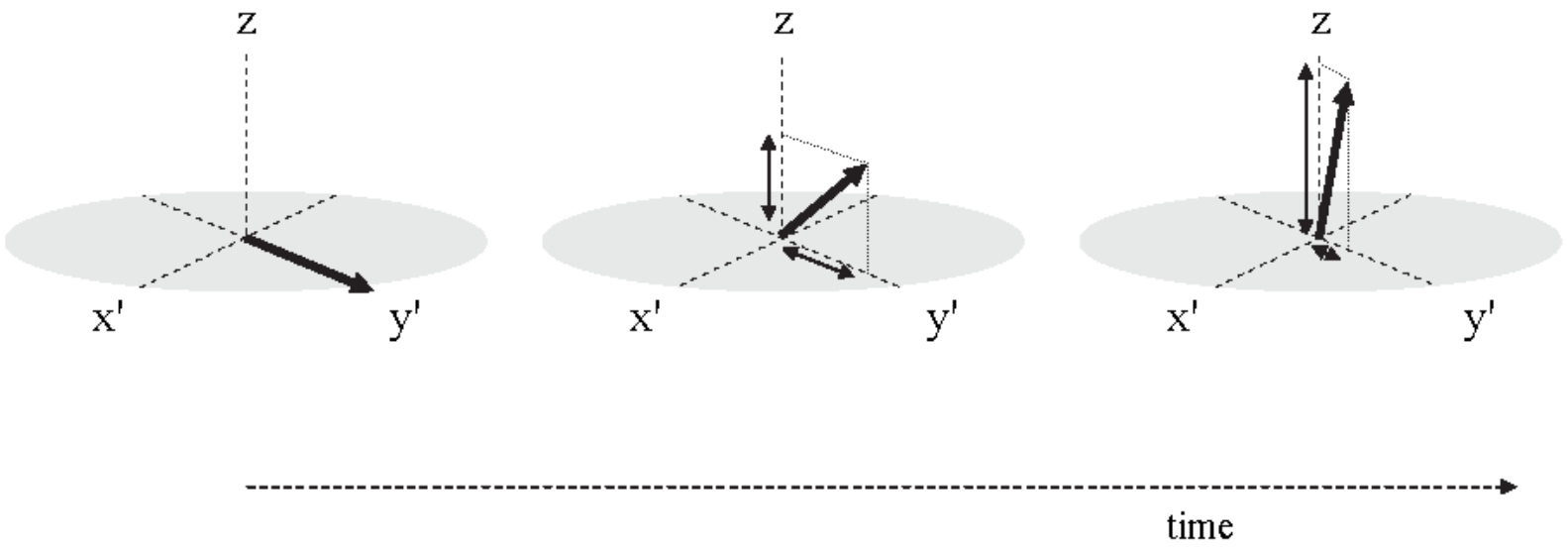


Relaxation & Relaxation Times

- At equilibrium magnetization state
  - z-component (M_z) equal to M_0
  - Transverse components (M_x and M_y) equal to zero
- Application of an RF pulse creates a non-equilibrium state
- After the pulse has been switched off, the system must relax back to the equilibrium
  - Magnetization components return in time to their values prior to the pulse being applied
  - Process is characterized by specific time-constants

Relaxation & Relaxation Times cont.

- Relaxation of the magnetization after application of a 90° pulse around the x axis



Relaxation & Relaxation Times cont.

- Two relaxation times govern the return to equilibrium
  - T_1 relaxation: affects only z-magnetization
  - T_2 relaxation: affects only xy-magnetization

Relaxation & Relaxation Times cont.

- T_1 time constant
  - Spin-lattice relaxation time: spins return energy to surrounding lattice
  - Related to the rate at which M_0 realign with B_0
  - After an RF pulse of arbitrary tip angle α, the value of M_z at a time t is given by:

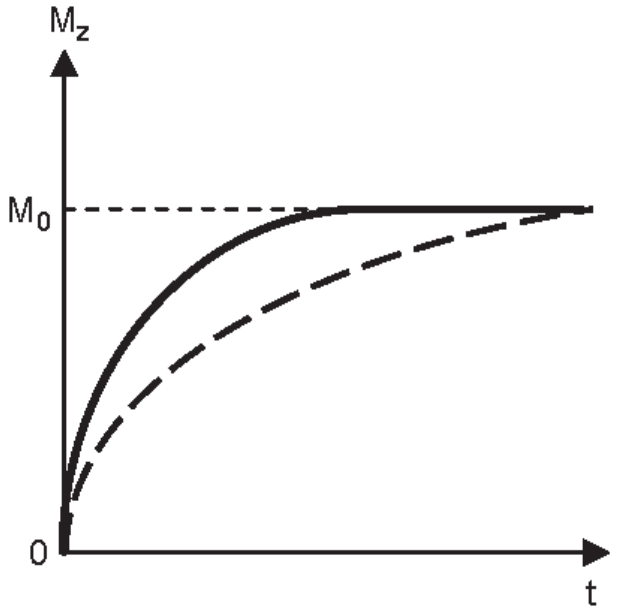
$$M_z(t) = M_0 \cos\alpha + (M_0 - M_0 \cos\alpha)\left(1 - e^{-\frac{t}{T_1}}\right)$$

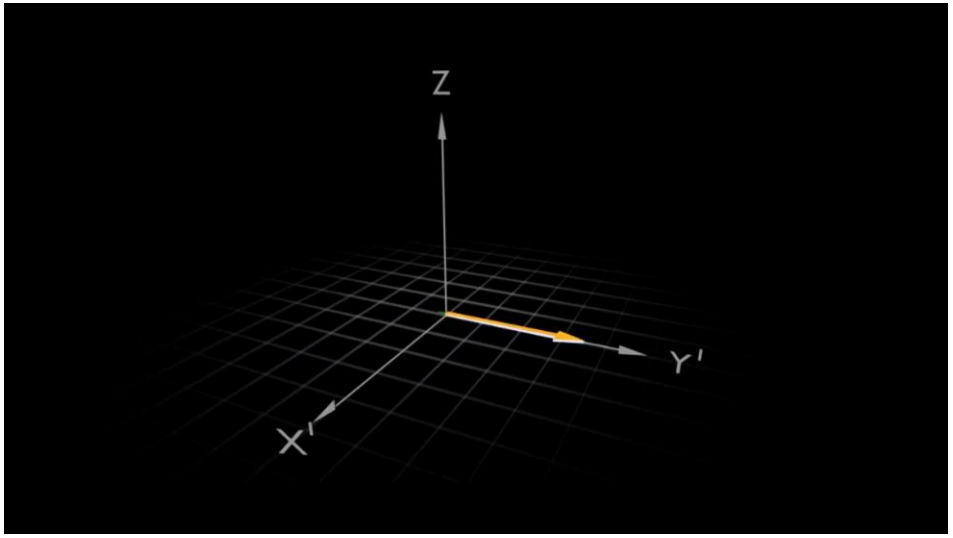
  - For a 90° pulse

$$M_z(t) = M_0\left(1 - e^{-\frac{t}{T_1}}\right)$$

Relaxation & Relaxation Times cont.

- T_1 time constant
  - Depends on the magnetic field strength
  - Is different for different tissue types
    - Solid line: a tissue with a short T_1
    - Dashed line: a tissue with a long T_1





Relaxation & Relaxation Times cont.

- T_2 time constant
  - Spin-spin relaxation time: spins lose coherence due to difference in magnetic environment
  - Related to rate at which M_x and M_y components return to the zero
  - After an RF pulse of arbitrary tip angle α, the value of M_y at a time t is given by:

$$M_y(t) = M_0 \sin\alpha \; e^{(\frac{-t}{T_2})}$$

  - For a 90° pulse

$$M_y(t) = M_0 \; e^{(\frac{-t}{T_2})}$$

Relaxation & Relaxation Times cont.

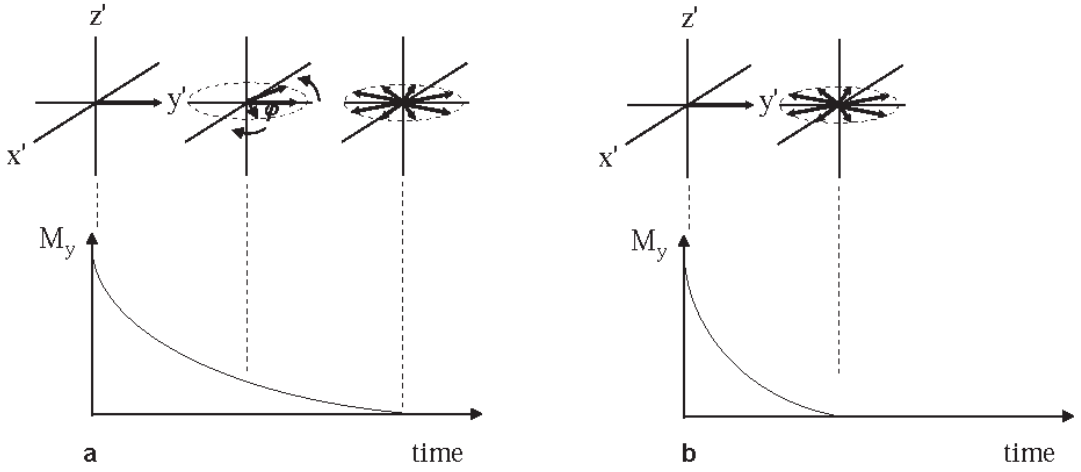
- T_2 time constant
  - Depends on the magnetic field strength and the tissue type
  - Accelerated by the spatial inhomogeneities in the B_0 field within the patient
    - Inhomogeneities in the B_0 field produced by the system
    - Local variations in magnetic field due to the different magnetic susceptibilities of different parts of the body

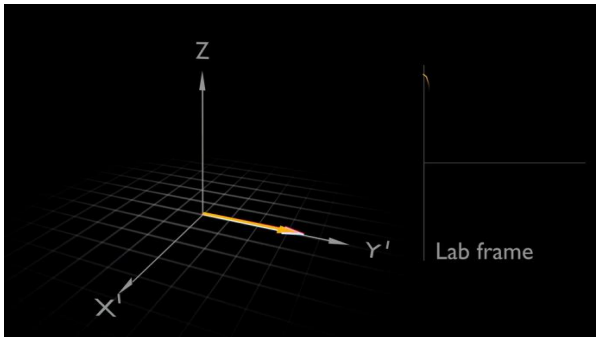
Relaxation & Relaxation Times cont.

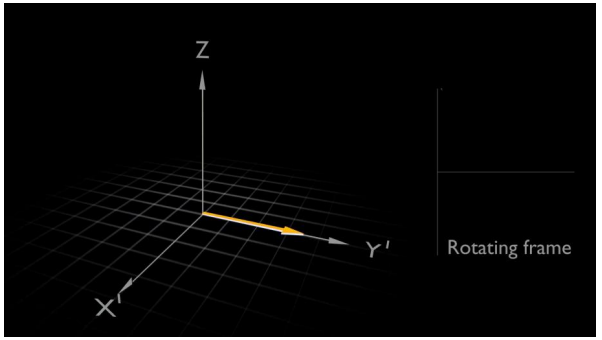
- T_2 time constant
  - There is a small spread in the precessing frequencies of protons
  - They accumulate different phase values and lose 'phase coherence'
  - The vector sum of all the individual magnetic moments decreases exponentially
  - When the vectors are randomly located around the transverse plane, M_x and M_y are relaxed back to zero
  - This process is known as "**dephasing**"

Relaxation & Relaxation Times cont.

- T_2 time constant
  - Example: The time-dependence of the M_y component of magnetization for a tissue with relatively (a) long and (b) shorter T2

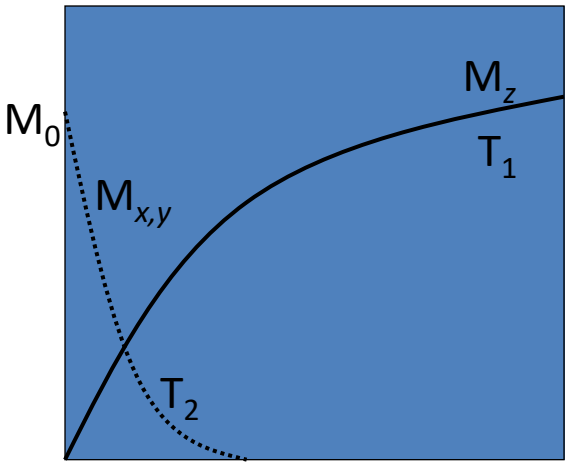






Relaxation & Relaxation Times cont.

- T_1 is always greater than T_2
  - But there is no direct correlation between the T_1 and T_2 values
- Tissue relaxation times (ms)



Tissue	T_1 (1.5 T)	T_1 (3 T)	T_2 (1.5 T)	T_2 (3 T)
Brain (white matter)	790	1100	90	60
Brain (grey matter)	920	1600	100	80
Liver	500	800	50	40
Skeletal muscle	870	1420	60	30
Lipid (subcutaneous)	290	360	160	130
Cartilage	1060	1240	42	37

Signals from Lipid

- Protons in lipid resonate at a significantly different frequency from those in water
  - Strength of the magnetic field experienced by a proton is affected by the geometry of the electron configuration which surrounds the proton

Signals from Lipid cont.

- Spinning electrons have an electron magnetic moment which produces a small magnetic field opposite in polarity to B_0 field
  - Reduces the effective magnetic field, B_{eff}, experienced by the proton

$$\mathrm{B_{eff}} = \mathrm{B_0}(1 - \sigma)$$

where σ is called the shielding constant, and is related to the electronic environment surrounding the nucleus

  - The resonant frequency of the proton is given by:

$$\omega = \gamma \mathrm{B_{eff}} = \gamma \mathrm{B_0}(1 - \sigma)$$

Signals from Lipid cont.

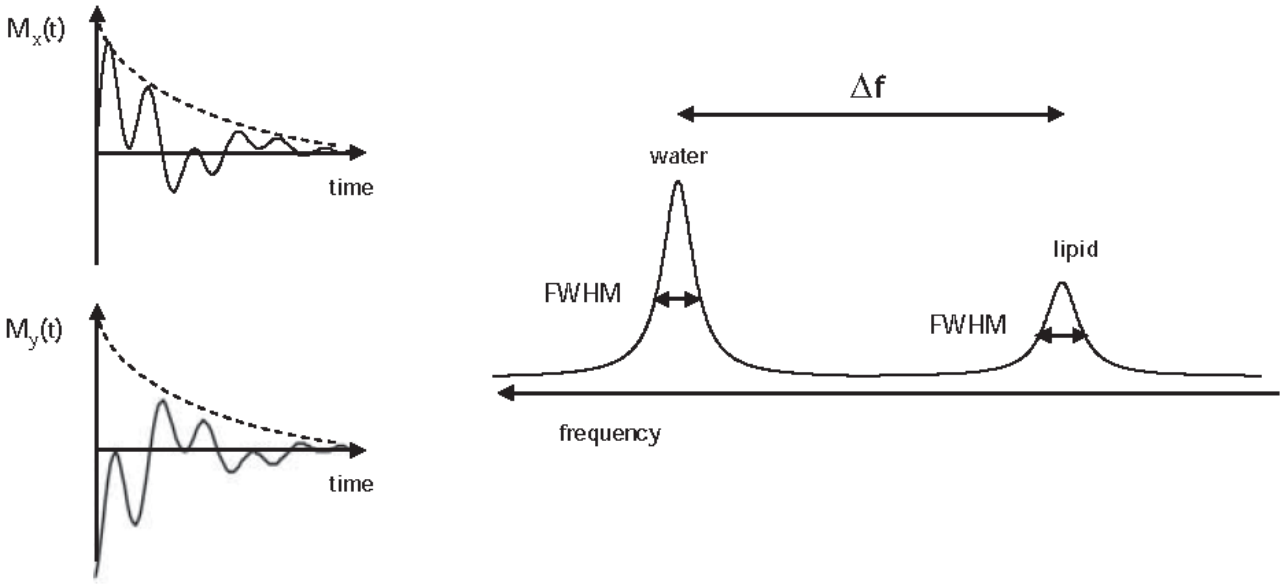
- Lipid ($-CH_2-$) and water (H_2O) have different electron distributions around the protons
  - Protons in the two species resonate at different frequencies
  - Value of σ for water is less than that for lipid, and the resonant frequency is higher
  - At 3 Tesla, the difference in resonance frequencies is roughly 2500 rad/s or 400 Hz

The Free Induction Decay (FID)

- The signal precesses freely after the RF pulse has been turned off
  - Detected via electromagnetic induction
  - Decays to a zero equilibrium value
- Combines the effects of an RF pulse, MR relaxation, chemical shift, and detection via Faraday induction

The Free Induction Decay (FID) cont.

- Example: the measured MR signal from a human subject
  - If two coils are present then both the M_x and M_y components of magnetization can be detected separately



Magnetic Resonance Imaging

- The FID signal contains no spatial information
  - No way to distinguish between signals coming from protons located at different spatial positions within the body
- If the magnetic field could be made to vary spatially within the subject
  - Spatial variation in resonant frequencies could be used to form an image
  - Variations should be dynamic
  - Performed by incorporating three separate ‘gradient coils’ in an MRI scanner

Magnetic Resonance Imaging

- Gradient coils are designed to have linear variation in magnetic field with respect to spatial location

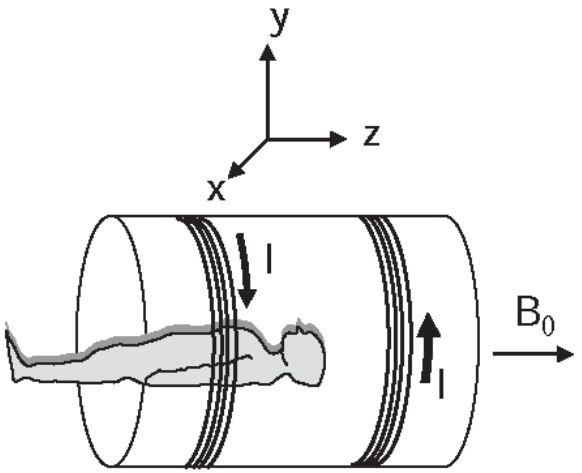
$$\frac{\partial \mathrm{B_z}}{\partial \mathrm{z}} = \mathrm{G_z}, \quad \frac{\partial \mathrm{B_z}}{\partial x} = \mathrm{G_x}, \quad \frac{\partial \mathrm{B_z}}{\partial y} = \mathrm{G_y}$$

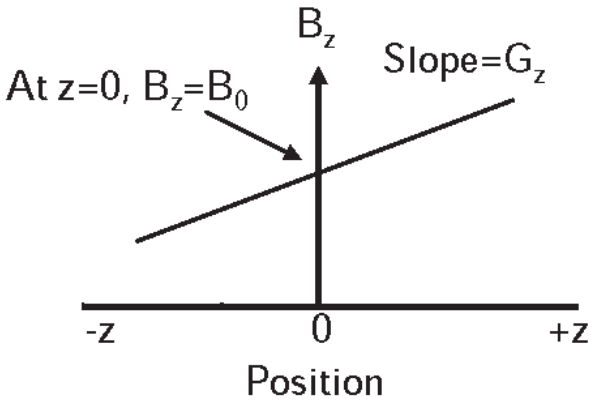
  where *G* represents the gradient measured in T/m

- Gradient coils does not additionally contribute to the magnetic field at the isocenter ($z = 0$, $y = 0$, $x = 0$) of the gradients
  - Magnetic field at this position is simply B_0

Magnetic Resonance Imaging cont.

- Example: a coil design to produce a linear gradient in the z-direction
  - A Maxwell-pair gradient coil in which an equal and opposite current is passed through a set of wires which are wound around a cylinder with its major axis in the *z*-direction





Magnetic Resonance Imaging cont.

- Currents in gradient coils
  - Comes from high power gradient amplifiers (hundreds of amps)
  - Can be turned on and off very quickly under computer control
  - Typical maximum values of gradient strength on clinical systems are ~40 mT/m
- The gradient coils are permanently fixed to the inside of the main magnet
- By convention (for a patient lying in the magnet)
  - *y*-axis: anterior/posterior direction
  - *x*-axis: left/right direction

Magnetic Resonance Imaging cont.

- The magnetic field, B_z, applied on protons with a common z-coordinate is given by:

$$\mathbf{B_z = B_0 + zG_z}$$

  - At position $z = 0$, $B_z = B_0$
  - For all positions $z < 0$, $B_z < B_0$,
  - For positions $z > 0$, $B_z > B_0$
- The precession frequencies (ω_z) of the protons, as a function of their position in z, are given by:

$$\omega_z = \gamma \mathbf{B_z} = \gamma(\mathbf{B_0 + zG_z})$$

Image Acquisition

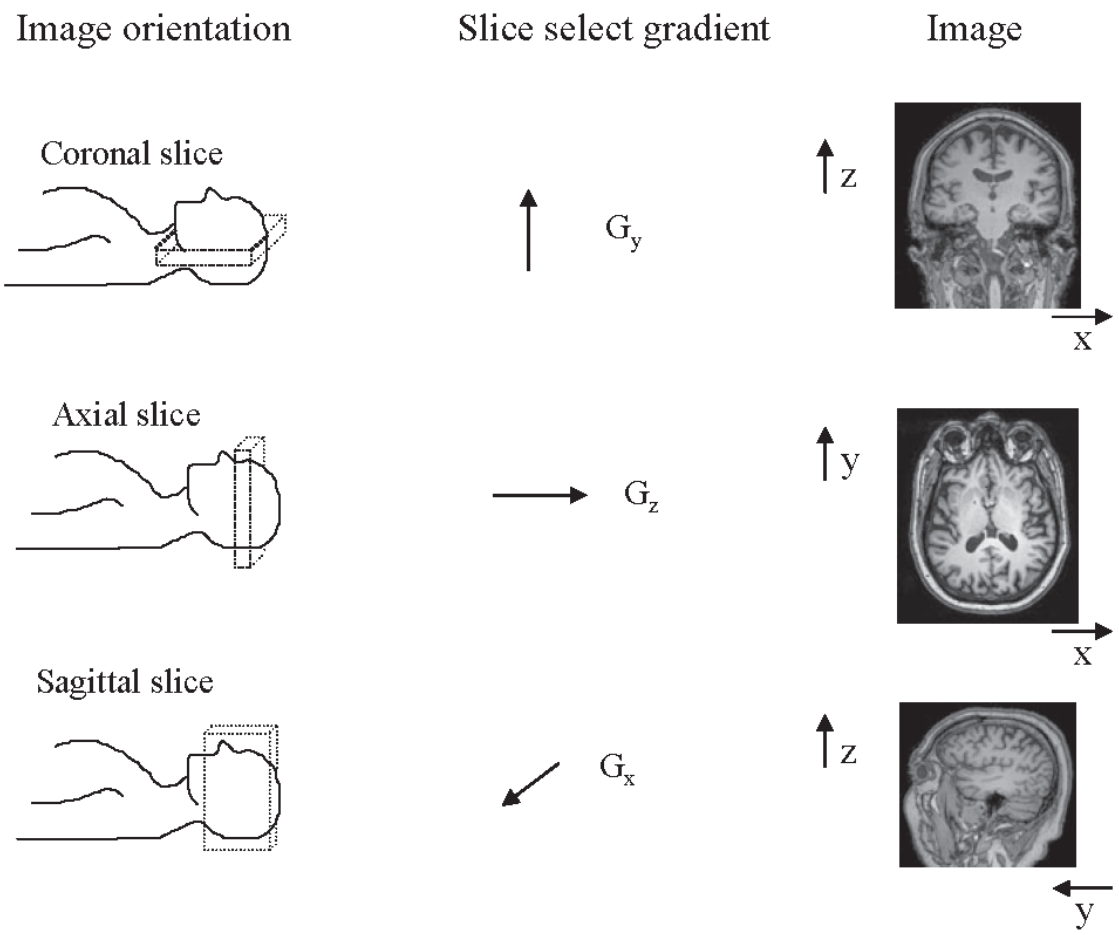
- Independent components of image formation
  - Slice selection
  - Phase-encoding
  - Frequency-encoding

Slice Selection

- MRI can acquire the image in any given orientation
  - Coronal, axial or sagittal slices (y-, z-, or x-directions)
  - Different clinical imaging protocols of various parts of the body call for different image orientations

Slice Selection cont.

- Different orientations for image acquisition

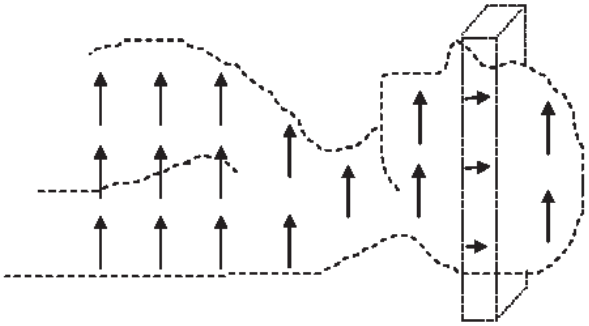


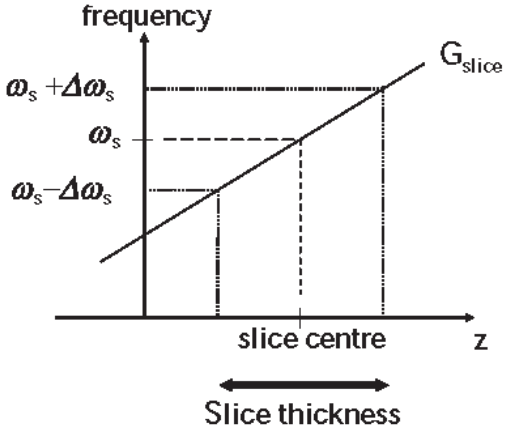
Slice Selection cont.

- Uses a frequency-selective RF pulse applied simultaneously with one of the magnetic field gradients (G_x, G_y or G_z), denoted by G_{slice}
  - At a specific frequency ω_s, with an excitation bandwidth of $\pm\Delta\omega_s$
  - Protons which have a precession frequency within the bandwidth are rotated into the transverse plane by the RF pulse
  - Protons with precession frequencies outside the bandwidth are not affected

Slice Selection cont.

- Orientation of the protons after a 90° pulse, assuming that an axial slice was chosen
  - Slice position selected by changing the value of ω_s of the RF pulse





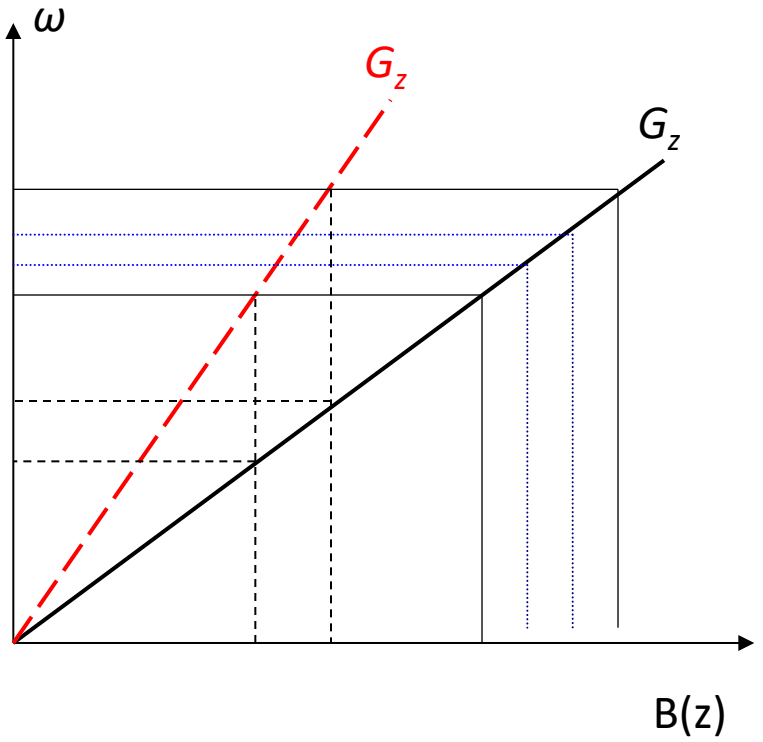
- The slice thickness (T) is given by:

$$T = \frac{2\Delta\omega_s}{\gamma G_{slice}}$$

Slice Selection cont.

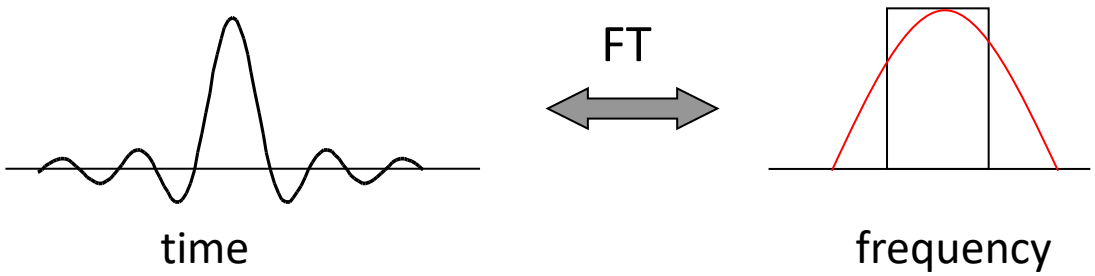
- Slice thickness is determined by the range of frequencies in the RF pulse and the gradient strength

- The slice position can be moved to different parts of the patient by changing the value of ω_s of the RF pulse



Slice Selection cont.

- Profile of exited region to be rectangle, $\pi\left(\omega/\Delta\omega\right)$
  - Slice-select RF pulse must be a sinc function, *sinc*($\Delta\omega t$)
  - To get perfect slice profile
    - RF slice must me infinitely long
    - In practice, truncated at a zero crossing (a few ms)



Slice Selection cont.

- RF pulse is relatively long
  - Protons at different z-positions within the slice accumulate different phases (φ_{slice})
  - A negative re-phasing gradient (G_{slice}^{ref}) is applied for a time τ^{ref} to 'reverse' this dephasing

$$\varphi_{\text{slice}}(z) = \gamma G_z z \frac{\tau}{2}$$

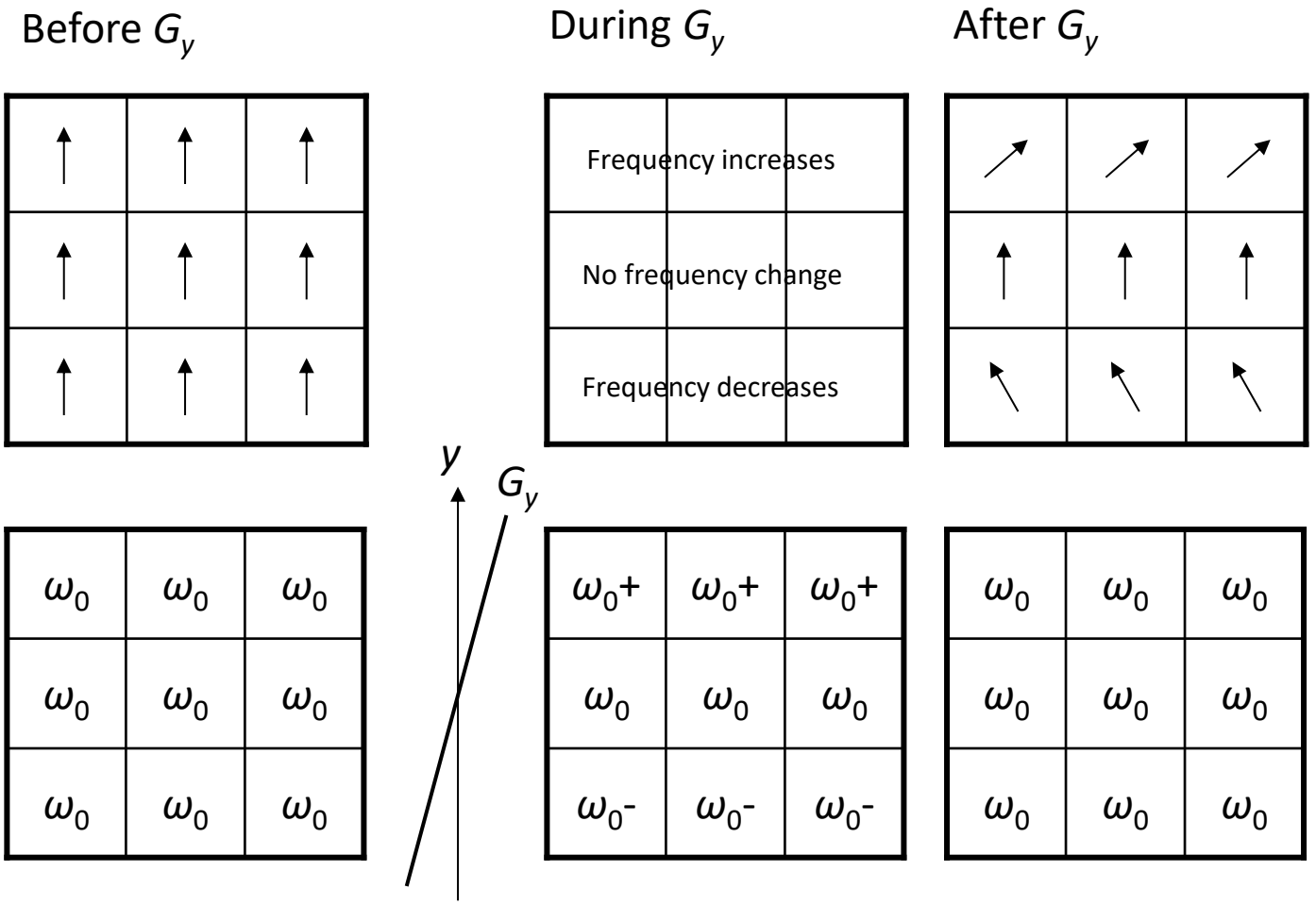
$$G_{\text{slice}}^{\text{ref}} \tau^{\text{ref}} = \frac{\tau}{2} G_z$$

Phase Encoding

- Continuing with the example of an axial slice
  - Slice select gradient was applied in the *z*-direction
  - *x* and *y* directions are now encoded via the phase and frequency of the MR signal – order doesn't matter
- A phase encoding gradient (G_{phase})
  - Turned on for a period τ_{pe} and then switched off
- If G_{phase} corresponds to the *y*-dimension
  - During the interval τ_{pe} the protons precess at a frequency $\omega_y = \gamma G_y y$
  - Introduce a spatially dependent phase shift:

$$\varphi_{\mathrm{pe}}\left(\mathrm{G_y}, \tau_{\mathrm{pe}}\right) = \omega_{\mathrm{y}}\tau_{\mathrm{pe}} = \gamma \mathrm{G_y y}\tau_{\mathrm{pe}}$$

Phase Encoding cont.



Frequency Encoding

- A frequency-encoding gradient (G_{freq})
  - While the receiver is on and data are being acquired
  - The x-dimension is encoded
  - A total of N_f data points are acquired
  - During this time t, protons precess at a frequency given by $\omega_x = \gamma G_x x$ determined only by their x-location

- Overall effect due to phase encoding, each voxel in the image is characterized by:
  - A specific *phase* depending on its y-position
  - A specific *frequency* depending on its x-position

Frequency Encoding cont.

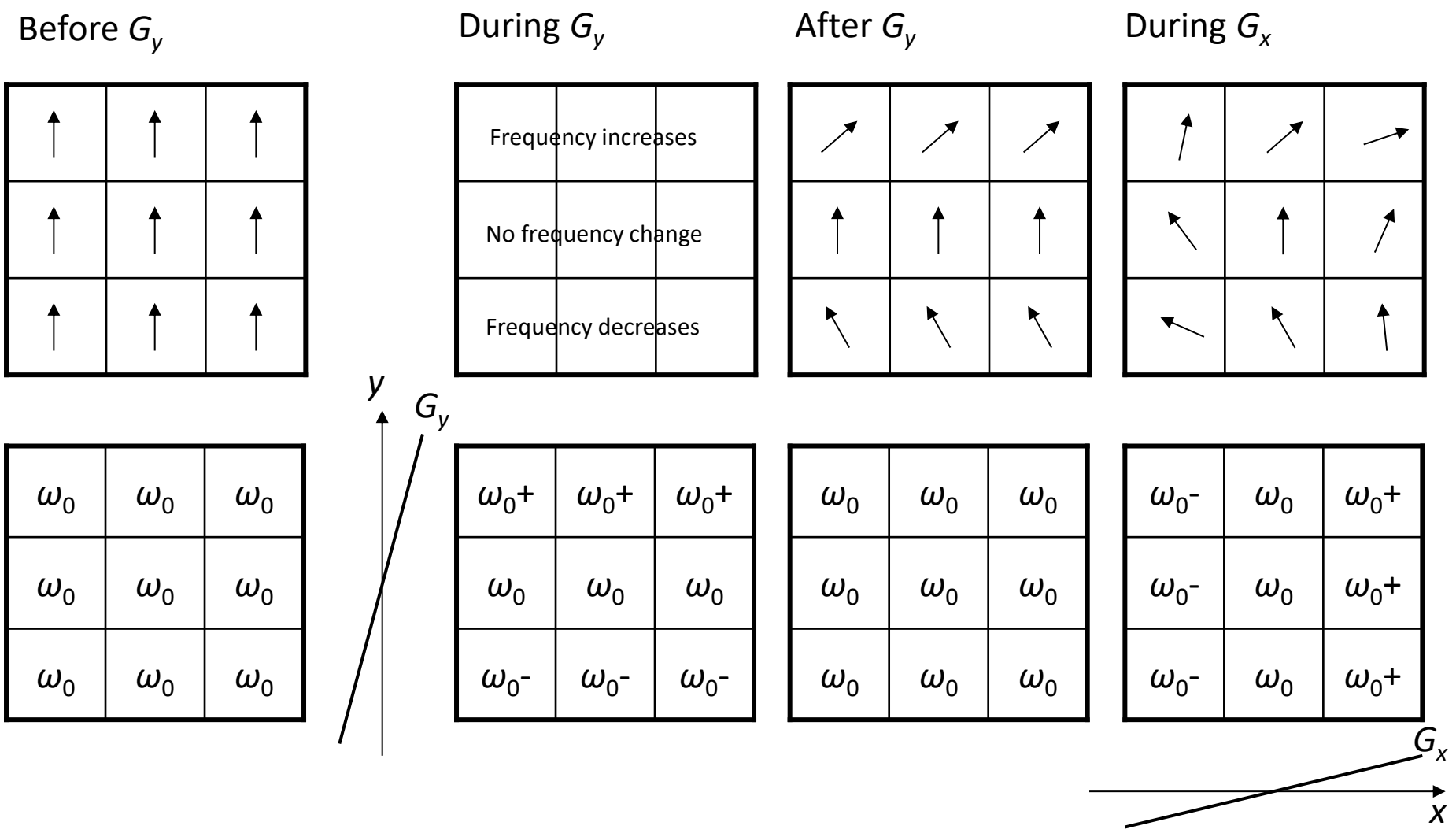
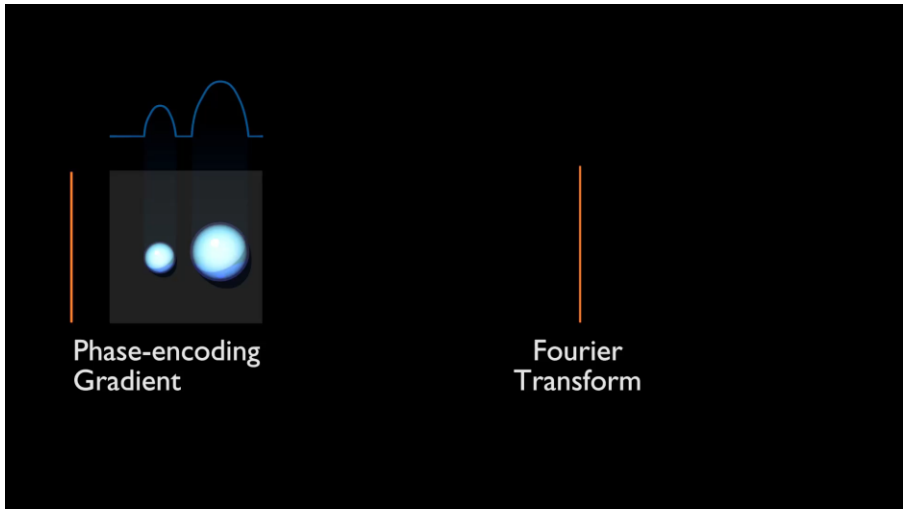


Image Formation

- Example: to form a 256 x 256 image
  - N_f = 256
  - Sequence must be repeated 256 times
    - Each time with a different value of the phase encoding gradient: maximum negative to maximum positive value in equal increments



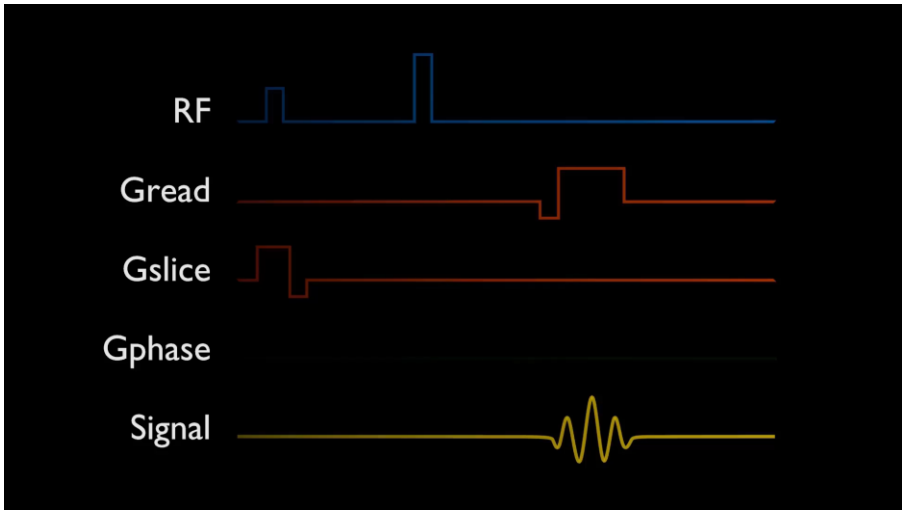


Image Formation cont.

- TR (time of repetition)
  - Delay between successive RF pulses
  - For sufficient T_1 relaxation to occur for a significant fraction of M_z magnetization to recover

- For a single slice acquisition:

  Total data acquisition time = TR.N_{pe}

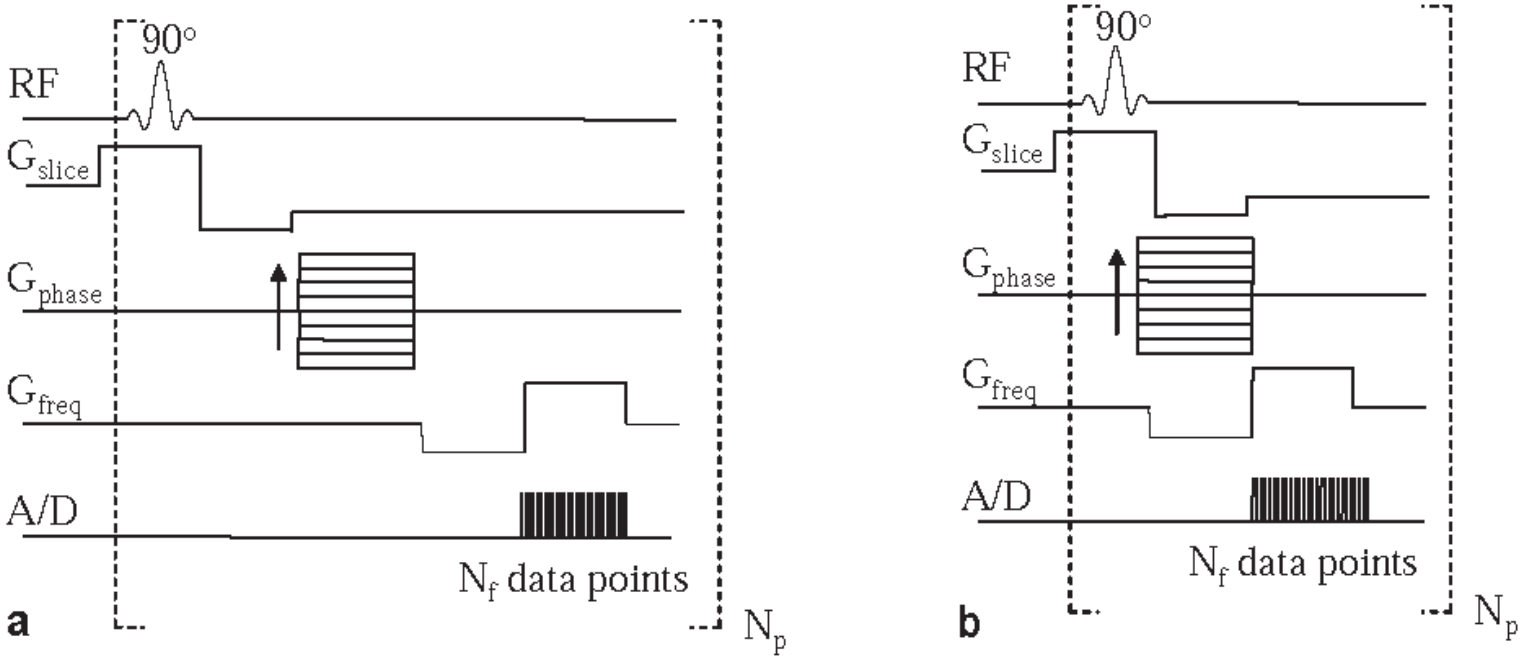
  Where N_{pe} = number of phase encoding steps applied

Pulse Sequence

- Pulse sequence diagrams for imaging sequences

  a: Individual steps in image formation

  b: In practice, the gradients are applied simultaneously where appropriate to minimize the time



Pulse Sequence cont.

- RF pulse
  - Length and power are adjusted to give the indicated tip angle
- Each gradient line
  - Height of the gradient pulse indicates the strength
  - Polarity (positive or negative) indicates the direction of current flow
- A/D line
  - Shows when the receiver is gated on
  - Indicates how many data points (N_f) are acquired

Pulse Sequence cont.

- N_p
  - Number of times the entire pulse sequence is repeated (typically between 128 and 512) to build-up a two-dimensional data set

- Arrow next to the phase encoding gradient
  - Indicates that different values are used for each repetition of the sequence

The k-space and Image Reconstruction

- The MRI signal for a selected slice is given by:

$$s\left(G_y, \tau_{pe}, G_x, t\right) \propto \int_{\text{slice}} \int_{\text{slice}} \rho(x, y) e^{-j\gamma G_y y \tau_{pe}} e^{-j\gamma G_x x t} dx dy$$

  - First term $\rho(x,y)$ is the number of protons at each position (x,y) within this slice (related to the number of protons in each voxel)
  - Second term represent the phase from phase encoding
  - Third term represent the frequency from frequency encoding

The k-space and Image Reconstruction cont.

- The k-space formalism
  - Two variables, k_x and k_y, are defined as:

$$k_x = \frac{\gamma}{2\pi} G_x t, \; k_y = \frac{\gamma}{2\pi} G_y \tau_{pe}$$

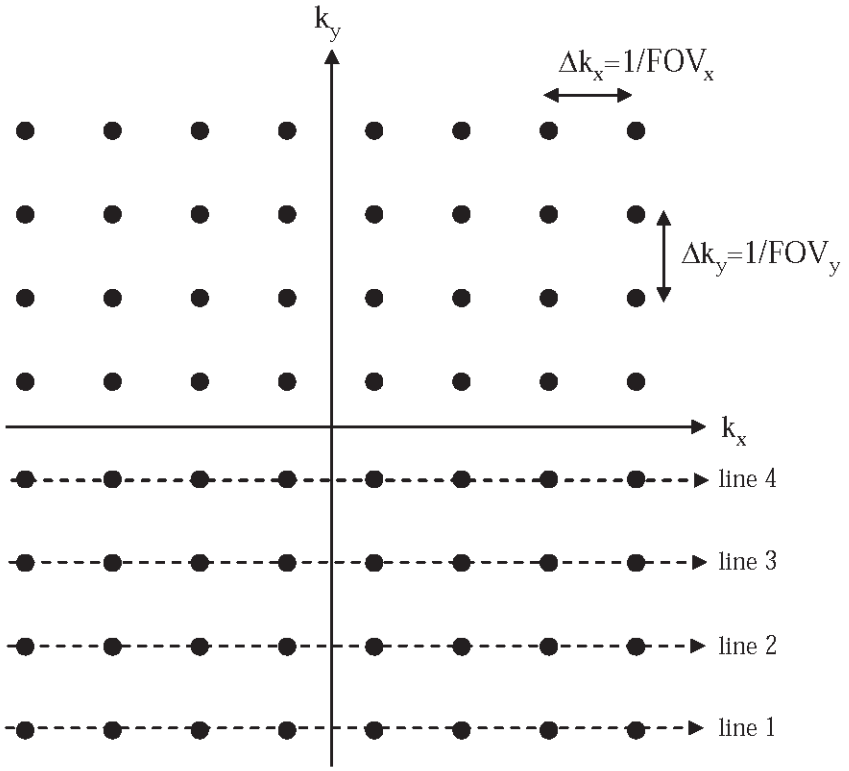
  - The *k* values are spatial frequencies
- The acquired signal is given by:

$$S(k_x, k_y) \propto \int_{\text{slice}} \int_{\text{slice}} \rho(x, y) e^{-j2\pi k_x x} e^{-j2\pi k_y y} dx dy$$

- Data is represented as a two-dimensional data set in k_x, k_y –space (*k*-space)

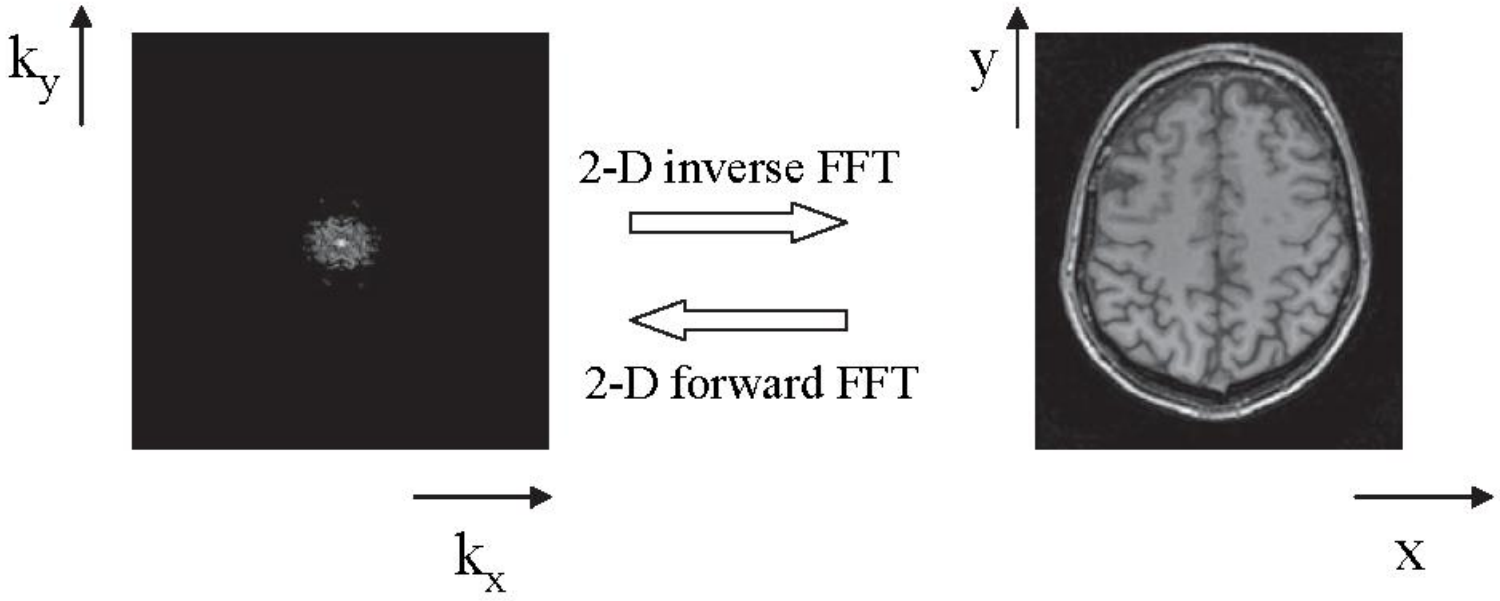
The k-space and Image Reconstruction cont.

- The k-space
  - Data acquisition occurs line-by-line
  - Δk_x and Δk_y between successive data points are equal to the inverse of the respective fields-of-view of the image

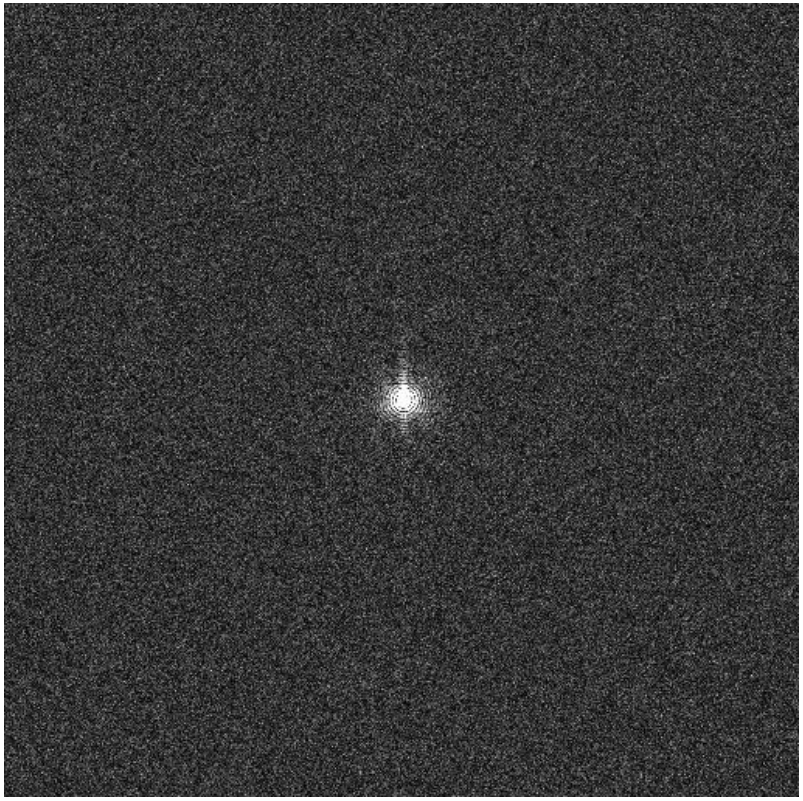


The k-space and Image Reconstruction cont.

- Inverse Fourier transform of a function $S(k_x,k_y)$
  - Gives $\rho(x,y)$ – MR image
  - Higher value of k_{max} , gives our higher spatial resolution in respective direction

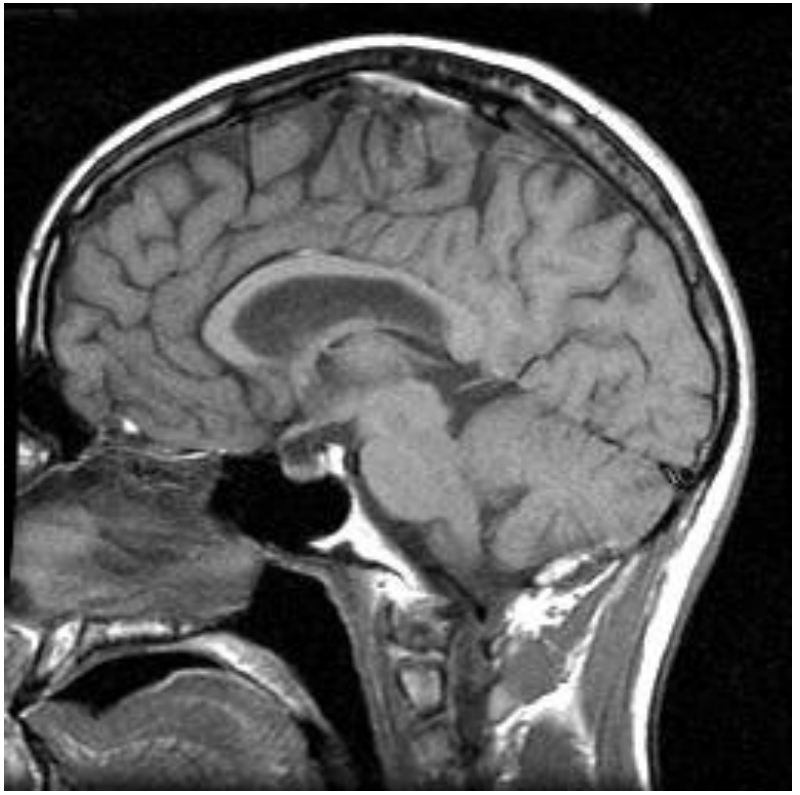


The k-space and Image Reconstruction cont.



k-space

FT



image

The k-space and Image Reconstruction cont.

- k-space is filled very fast in *x*-direction
  - Often 256 samples, $N_x = 256$
  - Sampling time = read time = A/D time
  - ~ few *ms*

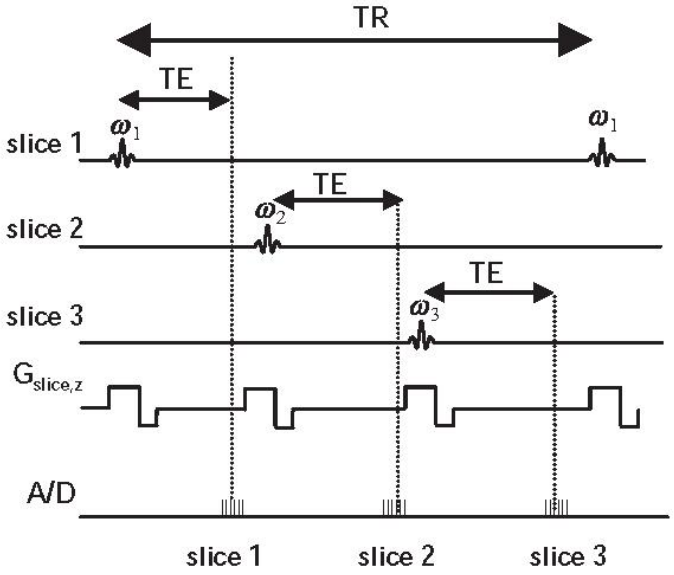
- Total scan time is limited by repetition time (TR)
  - Scan time = N_y . TR
  - If $N_y = 256$ and TR=2*s*, then scan time = 8.5 *minutes*

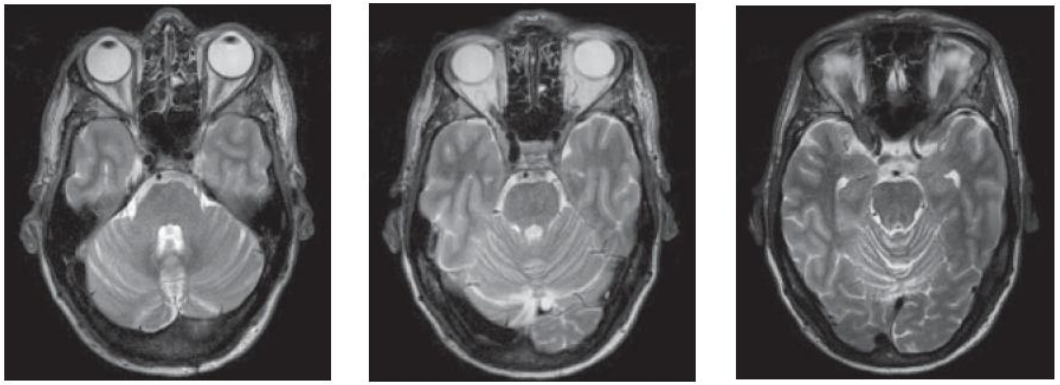
Multiple-slice Imaging

- If the normal process is repeated serially for every single slice MR scanning would take an extremely long time
  - T_2 values for tissue are much shorter than the T_1 relaxation
  - TR time required between successive RF excitations for each phase encoding step is much longer than the TE (Echo Time)
  - The ‘waiting time’ (TR – TE) can be used to acquire data from other slices

Multiple-slice Imaging cont.

- Multiple-slice gradient echo sequence





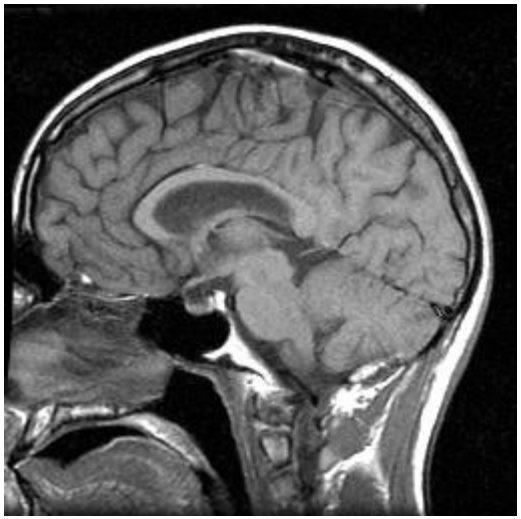
Multiple-slice Imaging cont.

- To excite a different slice
  - Only the centre frequency of the RF pulse is changed
  - Slice select, phase and frequency encoding gradients are repeated exactly as before
  - Data from this second slice are stored separately from the first
  - Slices are acquired in an interlaced fashion
    - All the odd-numbered slices followed by all the even-numbered
    - The maximum number of slices is given by the value of TR/TE

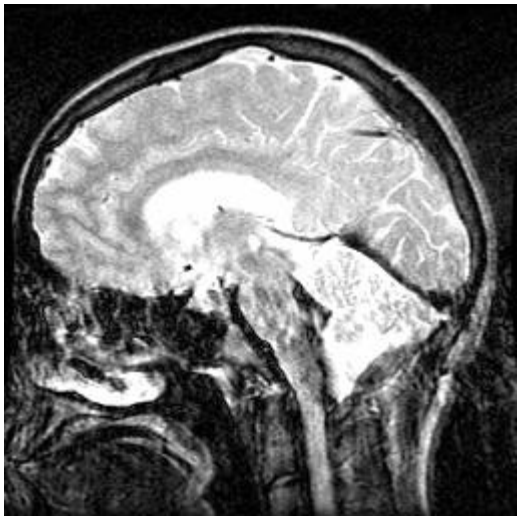
Contrast Manipulation

- MRI signal depends on
  - Number of “mobile” protons
  - T_1 time constant
  - T_2 time constant
- Proton density weighted (PDW) imaging
  - Minimizes signal differences due to T_1 and T_2
- T_1 weighted (T1W) imaging
  - Minimizes signal differences due to T_2
- T_2 weighted (T2W) imaging
  - Minimizes signal differences due to T_1

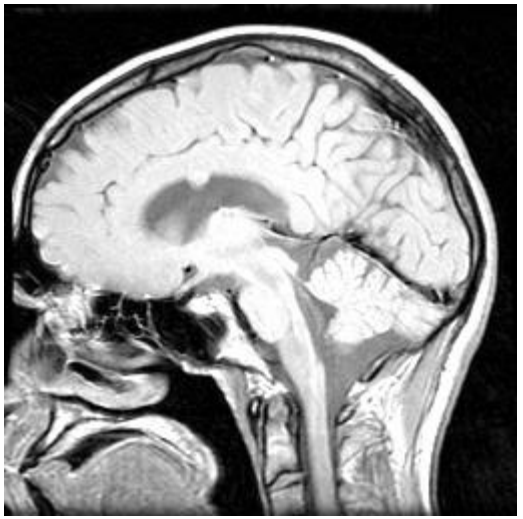
Contrast Manipulation cont.



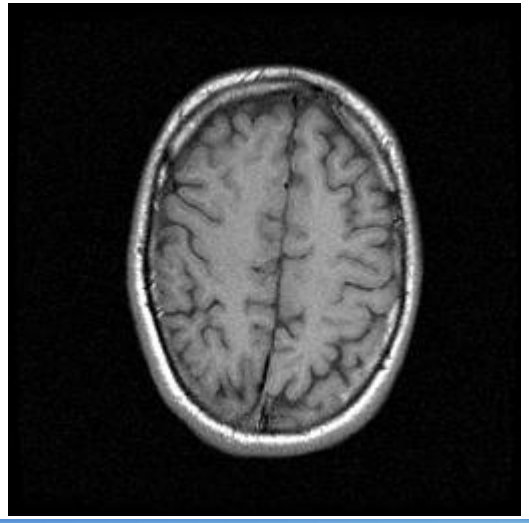
T1W

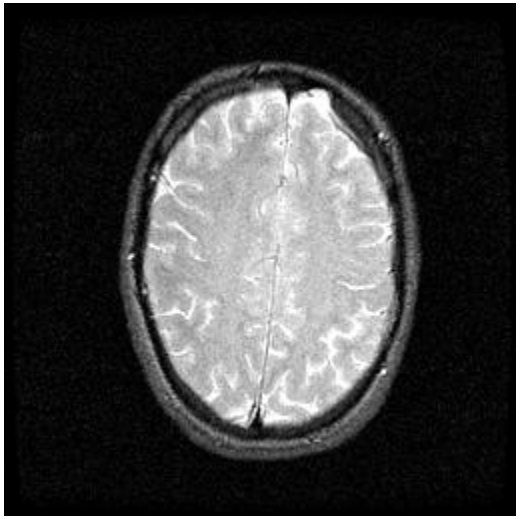


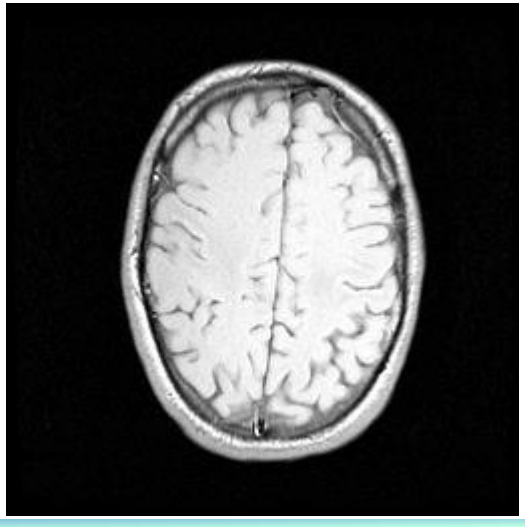
T2W



PDW







Imaging Sequences

- A typical clinical MRI protocol
  - Takes between half-an-hour and an hour
  - May consist of between three and six different types of scans
- Basic imaging sequences
  - Multi-slice gradient echo sequences
  - Spin echo sequences
  - Three-dimensional imaging sequences
- Fast imaging sequences
  - Echo planar imaging
  - Turbo spin echo sequences

Multi-slice Gradient Echo Sequences

- Basic multi-slice imaging sequence
  - Usually needs a long TR to allow full T1 relaxation to occur at 90° tip angles
  - Gives rise to a long image acquisition time
  - To image more rapidly, tip angle (α) is reduced to a value considerably smaller than 90°
  - Image intensity of pixel, I(x,y), is given by:

$$I(x,y) \propto \rho(x,y) \frac{\left(1 - e^{-\frac{TR}{T_1}}\right) \sin\alpha}{1 - e^{-\frac{TR}{T_1}} \cos\alpha} e^{-\frac{TE}{T_2^*}}$$

  - Optimized tip angle α_{Ernst} for a given TR for maximum SNR:

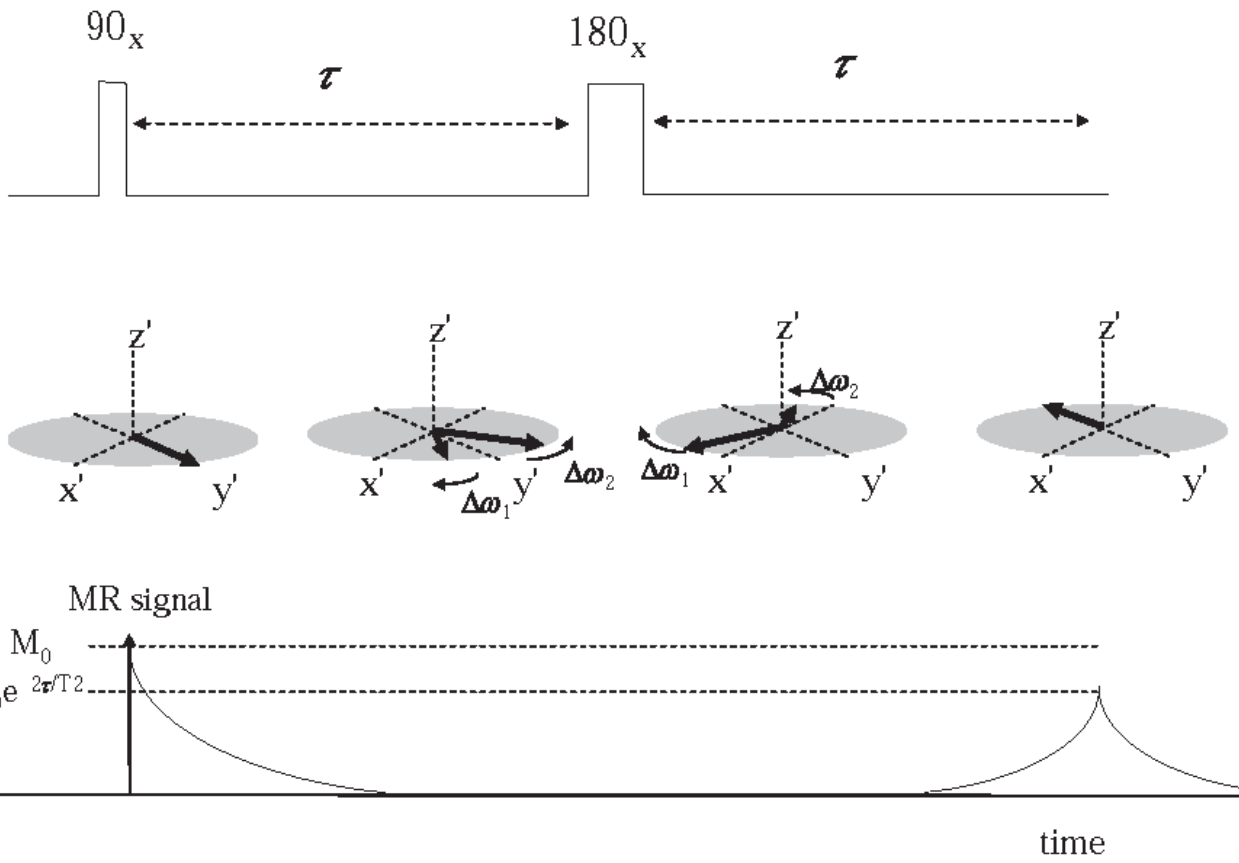
$$\alpha_{Ernst} = \cos^{-1} e^{-\frac{TR}{T_1}}$$

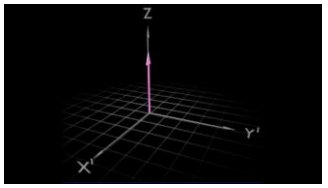
Spin Echo Sequences

- Most frequently used pulse sequence
  - T_2 values in gradient echo sequences are affected by magnet and tissue inhomogeneities
  - A second 180° pulse refocuses the effects of inhomogeneities
  - The intensity of an axial image is given by:

$$I(x,y) \propto \rho(x,y)\left(1 - e^{-\frac{TR}{T_1}}\right)e^{-\frac{TE}{T_2}}$$

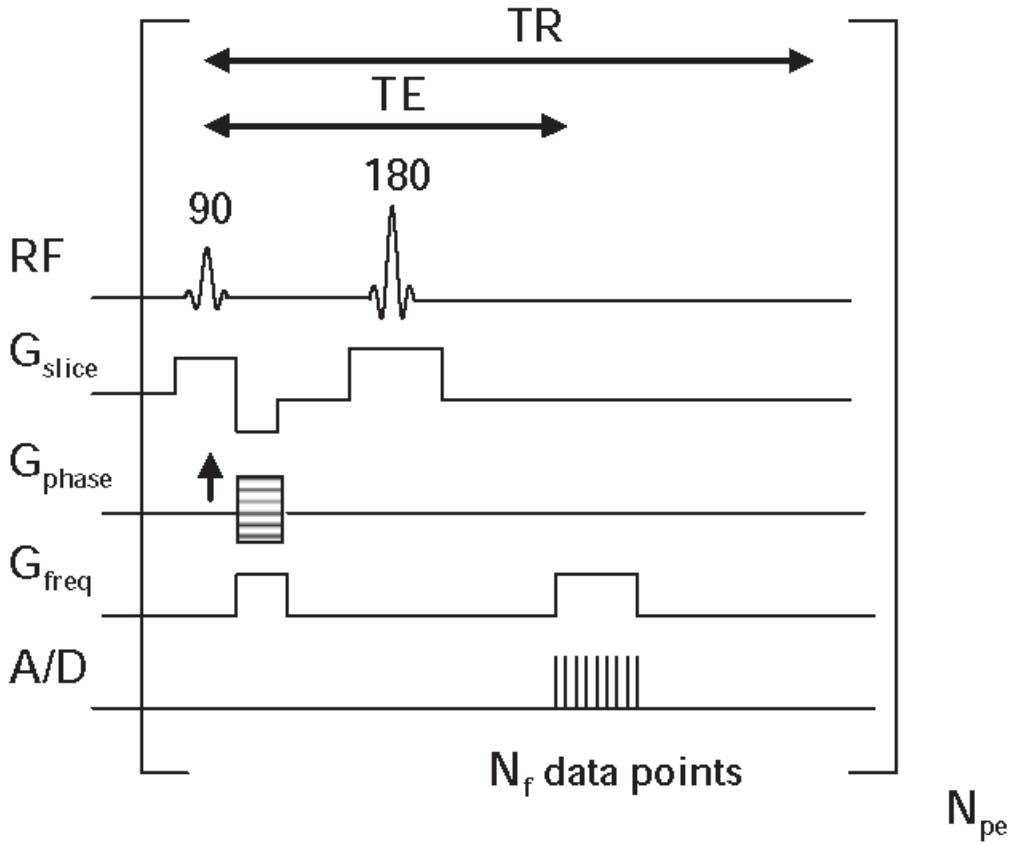
Spin Echo Sequences cont.





Spin Echo Sequences cont.

- Basic pulse sequence

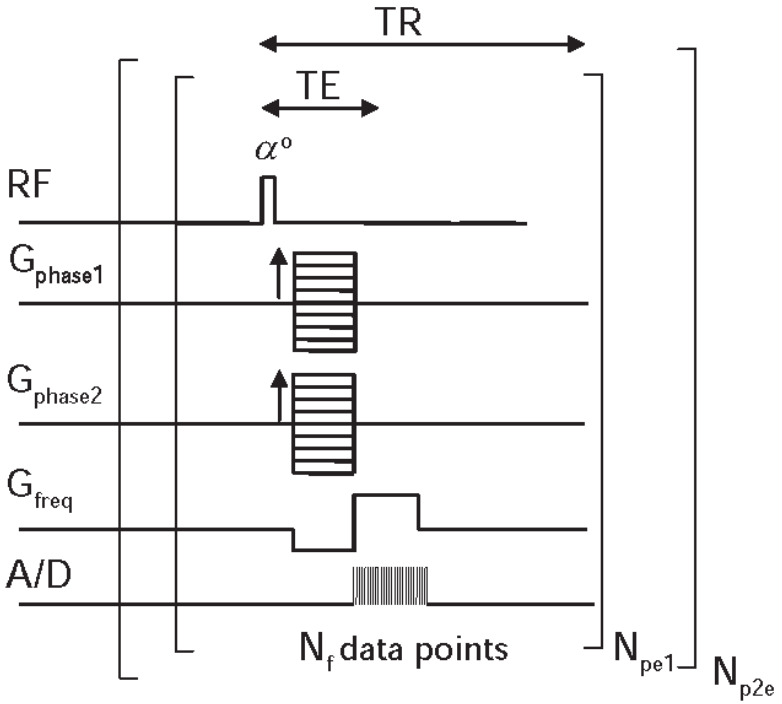


Three-dimensional Imaging Sequences

- Multi-slice imaging sequences
  - In-plane image resolution (1 mm or less) is typically much higher than the slice thickness (2-3 mm)
- A three-dimensional sequence
  - Isotropic high spatial resolution
  - There are two incremental phase encoding gradients and one frequency encoding gradient
  - TR must be very short → most 3D sequences are gradient-echo based

Three-dimensional Imaging Sequences cont.

- Three-dimensional gradient echo sequence



$$S(k_x, k_y, k_z) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \rho(x, y, z) e^{-j2\pi(k_x x + k_y y + k_z z)} dk_x dk_y dk_z$$

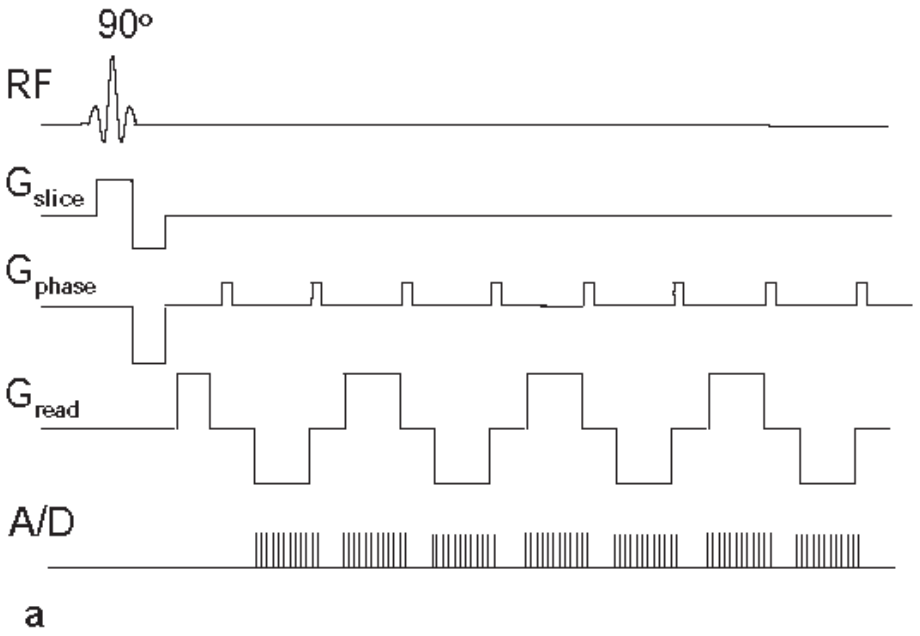
Echo Planar Imaging

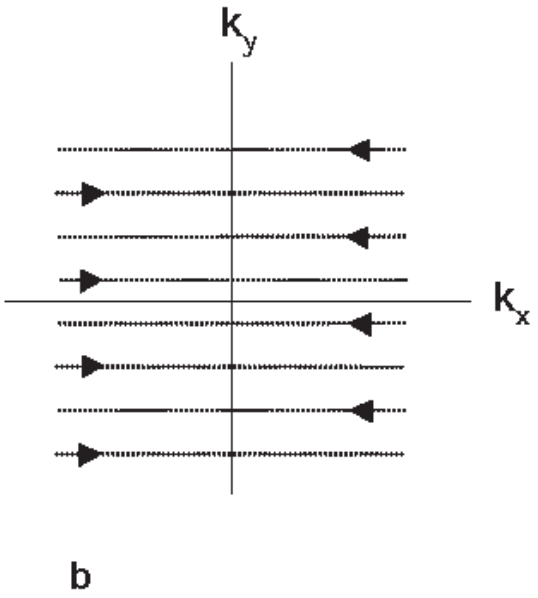
- For a small number of applications in which faster imaging is required
  - Diffusion tensor imaging
  - Functional MR imaging
- Acquires the image from each slice sequentially
  - No conventional TR delay
  - Each slice can be acquired in ~50–100 ms
- Major disadvantage
  - Signal is weighted by T_2 which can cause significant image distortions

Echo Planar Imaging

a: Basic echo planar imaging sequence used for single-shot rapid MRI

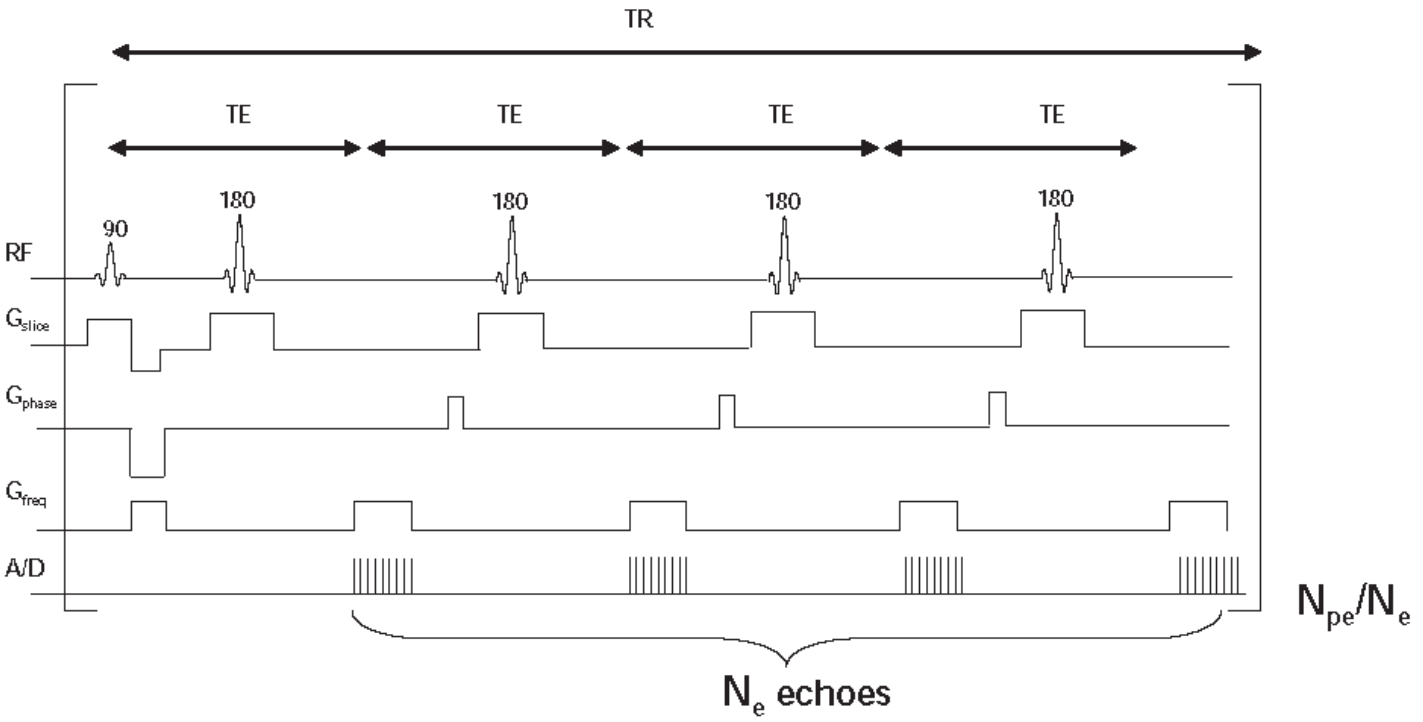
b: Corresponding k-space coverage





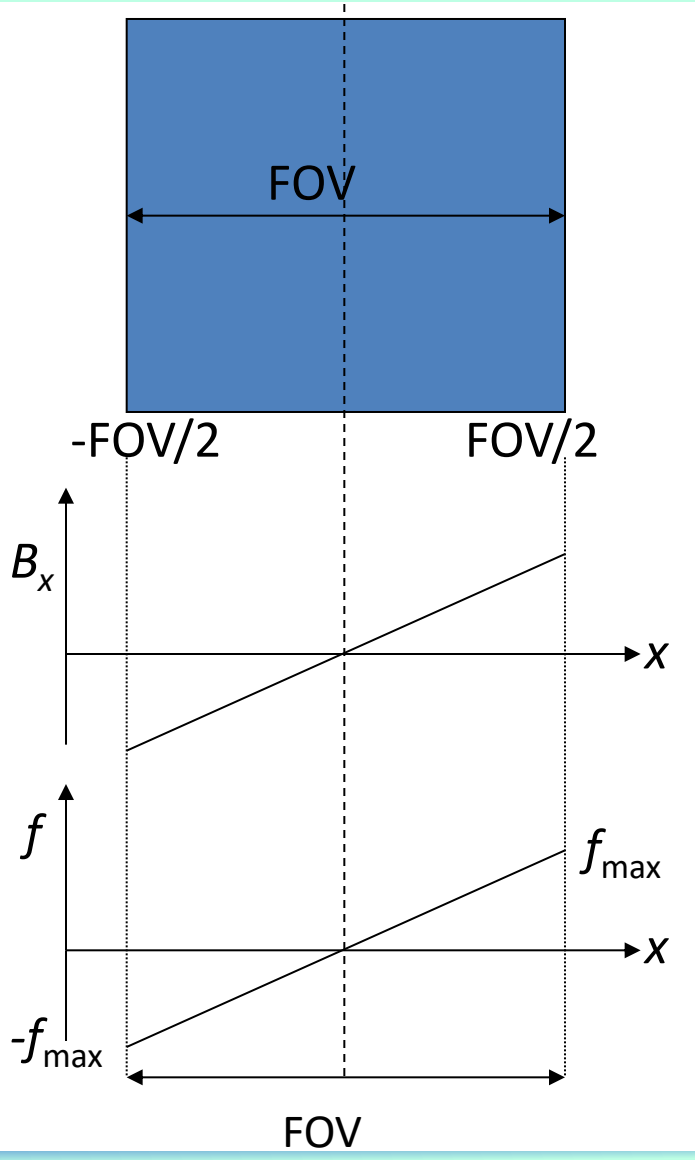
Turbo Spin Echo Sequences

- Multiple spin echoes in one TR to achieve a high imaging speed
  - No distortions due to T_2 weighting

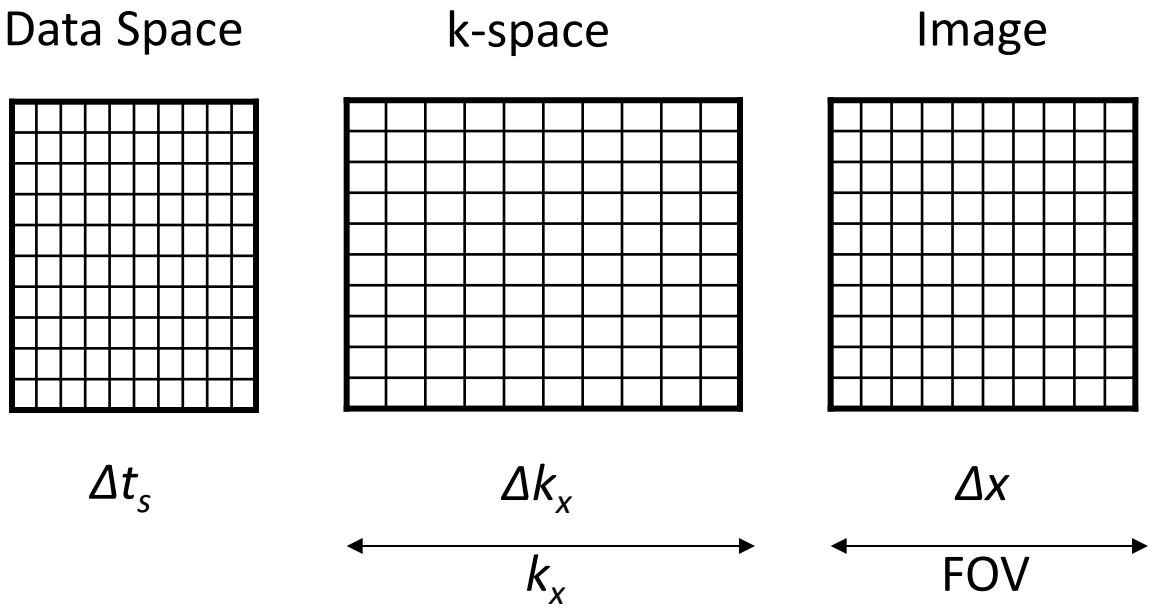


Field of View: *x*-direction

- FOV depends on the gradient strength and the range of frequencies



Field of View: *x*-direction cont.



$$FOV = \frac{1}{\Delta k_x}$$

$$\Delta x = \frac{1}{k_x}$$

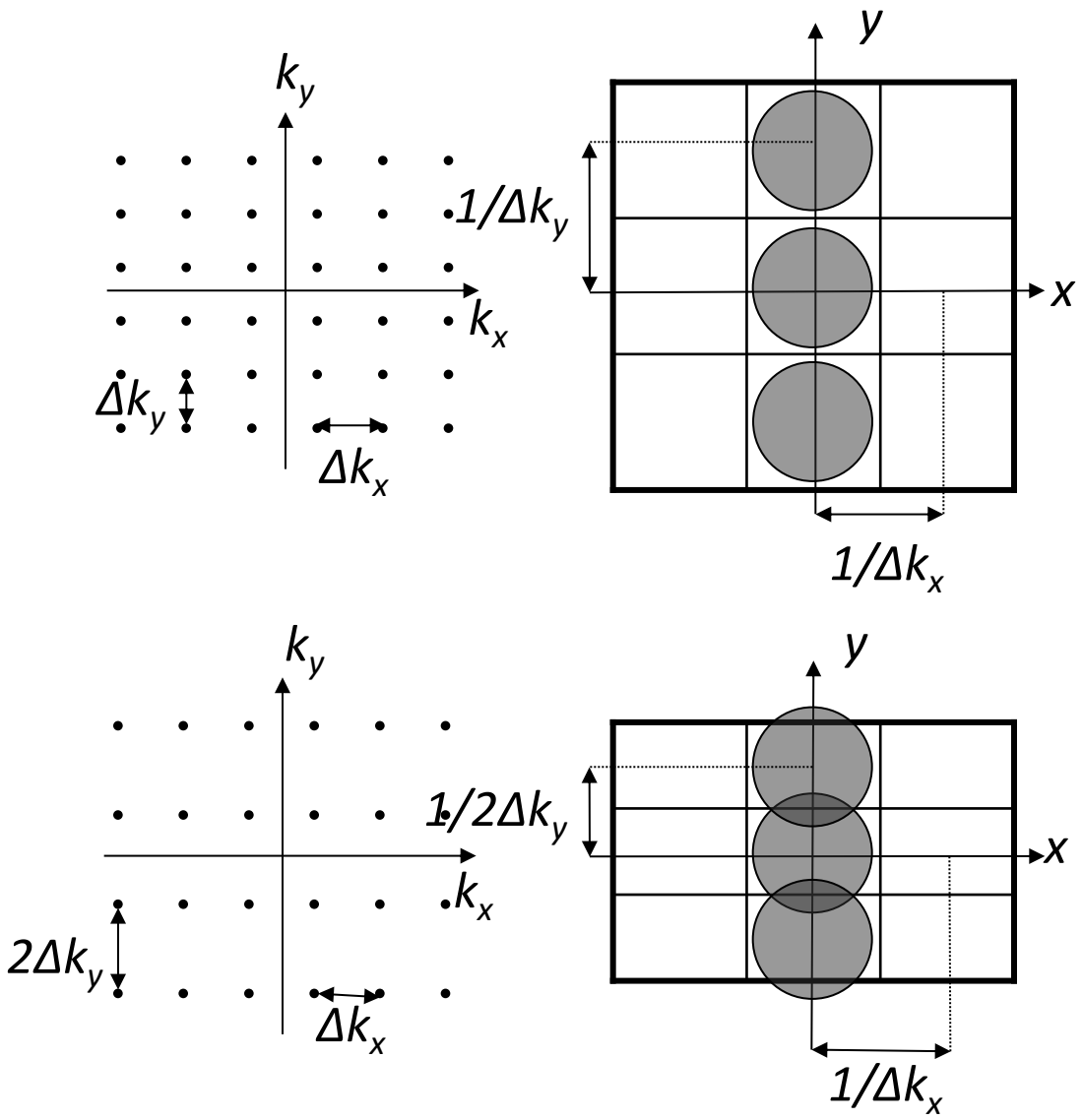
- Smaller Δk_x → larger FOV
- Smaller k_x → larger pixel size Δx (poor resolution)

Field of View: *y*-direction

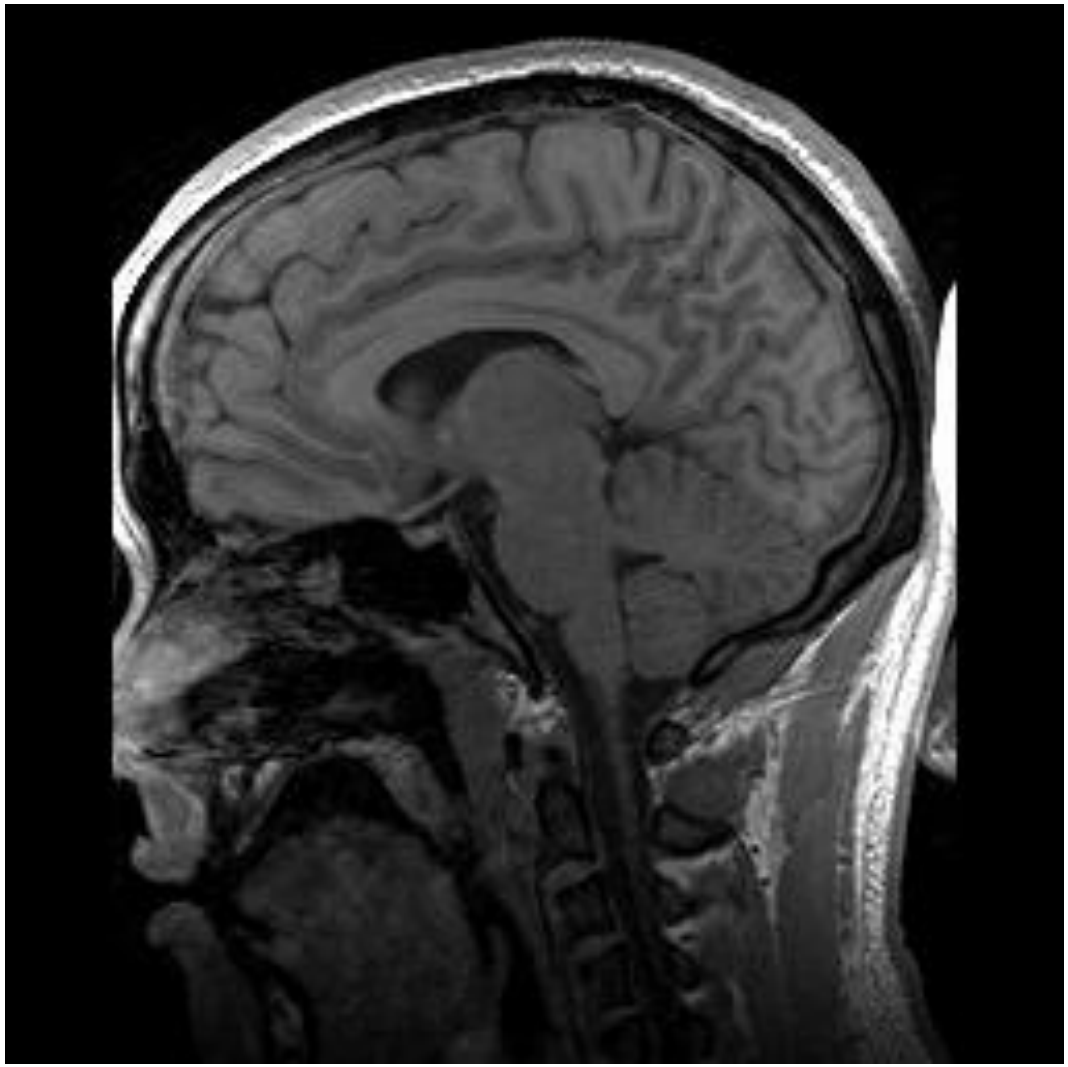
- One line of k-space is acquired for every TR
- A different G_y strength is used for every k_y line (phase encoding)
  - In the *y*-direction the sampling interval Δk_y depends on the incremental area of the phase encoding gradient G_y (i.e. $\Delta G_y \Delta T$)

Aliasing

- FOV cannot be reduced at will
  - Due to the sampling nature of MRI acquisition



Aliasing cont.



Signal-to-Noise Ratio (SNR)

- SNR in MR imaging depends on
  - Voxel volume
  - Total data acquisition time (time during which A/D converters are ON)

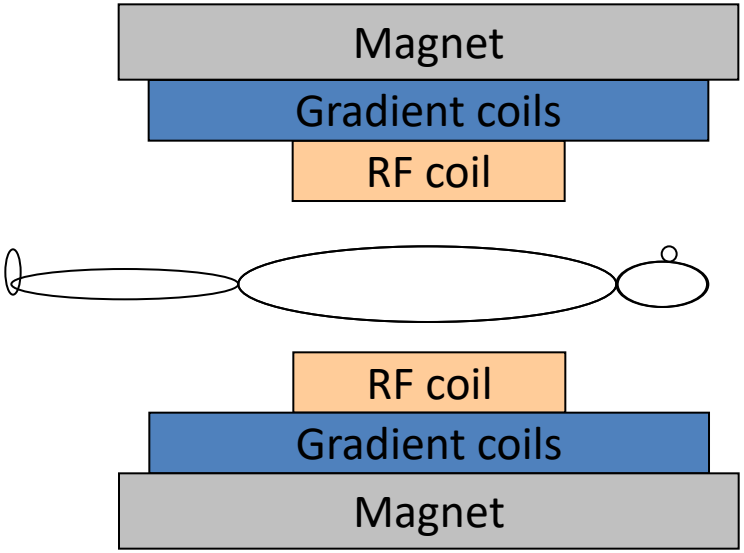
$$\text{SNR} \propto \frac{\Delta x \Delta y \Delta z \sqrt{N_x N_y \text{NEX}}}{\sqrt{\text{BW}}}$$

N_x = no.of samples in x
N_y = no.of samples in y
NEX = number of excitations
BW = signal bandwidth = $1/\Delta t_s$

$$\text{SNR} \propto \Delta x \Delta y \Delta z \sqrt{(N_x \Delta t_s) N_y \text{NEX}}$$

$$\text{SNR} \propto \text{voxel volume} \sqrt{\text{total A/D time}}$$

MRI Instrumentation





Main Magnetic Field, B_0

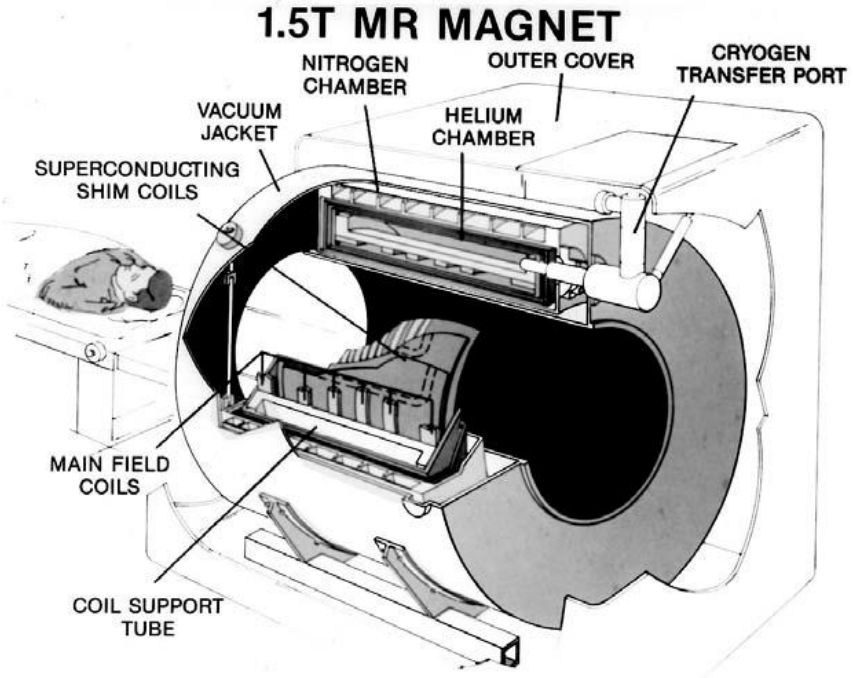
- Requirements of an imaging magnets are:
  - High magnetic field strength
  - Uniform magnetic field
    - Homogeneity is specified over a given volume (eg. 5 *ppm* over 50 cm^3
  - Temporal stability
    - No drift in magnetic field over time
    - Usually not an issue with superconducting materials
  - Patient access to imaging volume
  - Small extent of fringe field

Magnets

- Permanent
  - Max possible field strength ~0.3 *T*
  - “Open Magnet” design
- Resistive
  - Magnetic field produced from an electric current running through a coil
  - High power consumption
  - Sensitive to power fluctuations
  - Need to be cooled
  - Low field strength

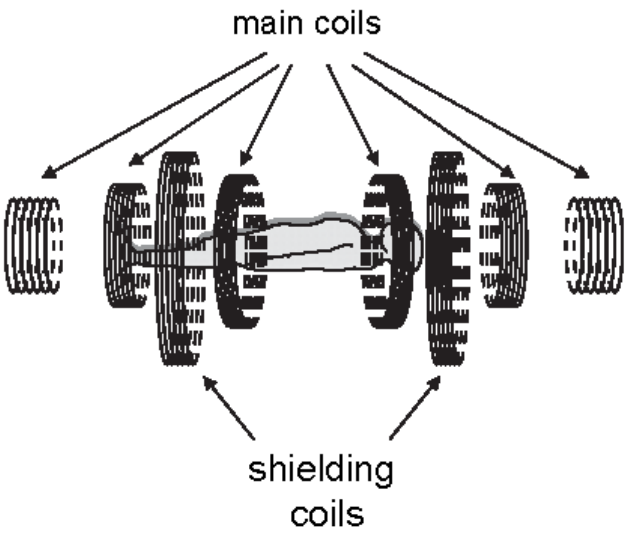
Magnets cont.

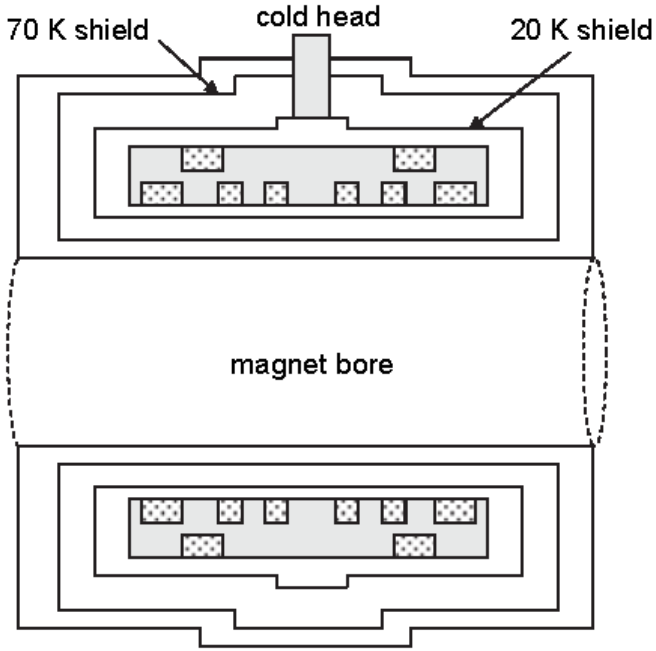
- Superconducting
  - Also electromagnets
  - Resistance drops to zero at 4K (liquid He)
    - Once current starts flowing it will continue to flow
  - Made of precisely wound coils
    - To control homogeneity
  - Mostly used in modern MRI



Magnets cont.

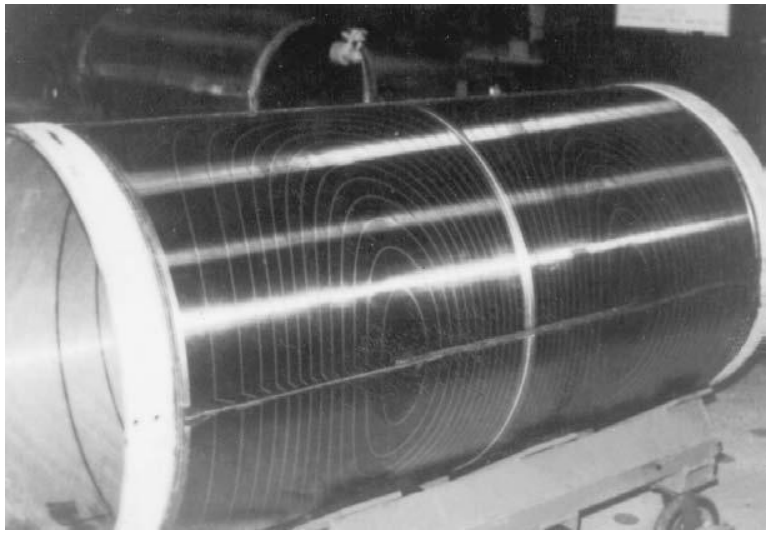
- Superconducting magnet design
  - Solenoidal coils (six) used to produce a homogeneous static magnetic field
  - Shielding coils (two) are used to reduce the effect of the 'stray field'

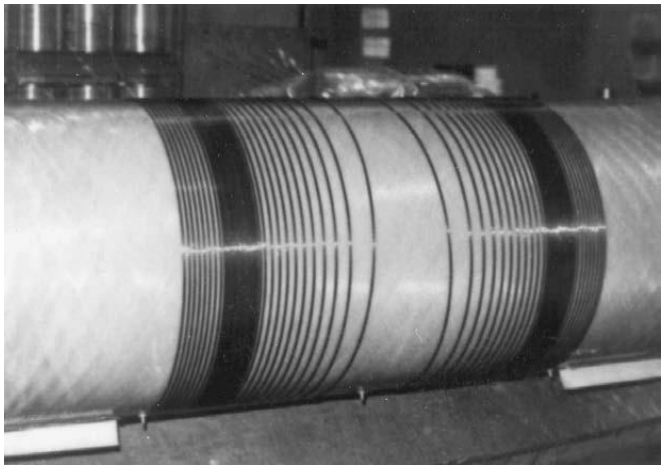




Gradient Coils

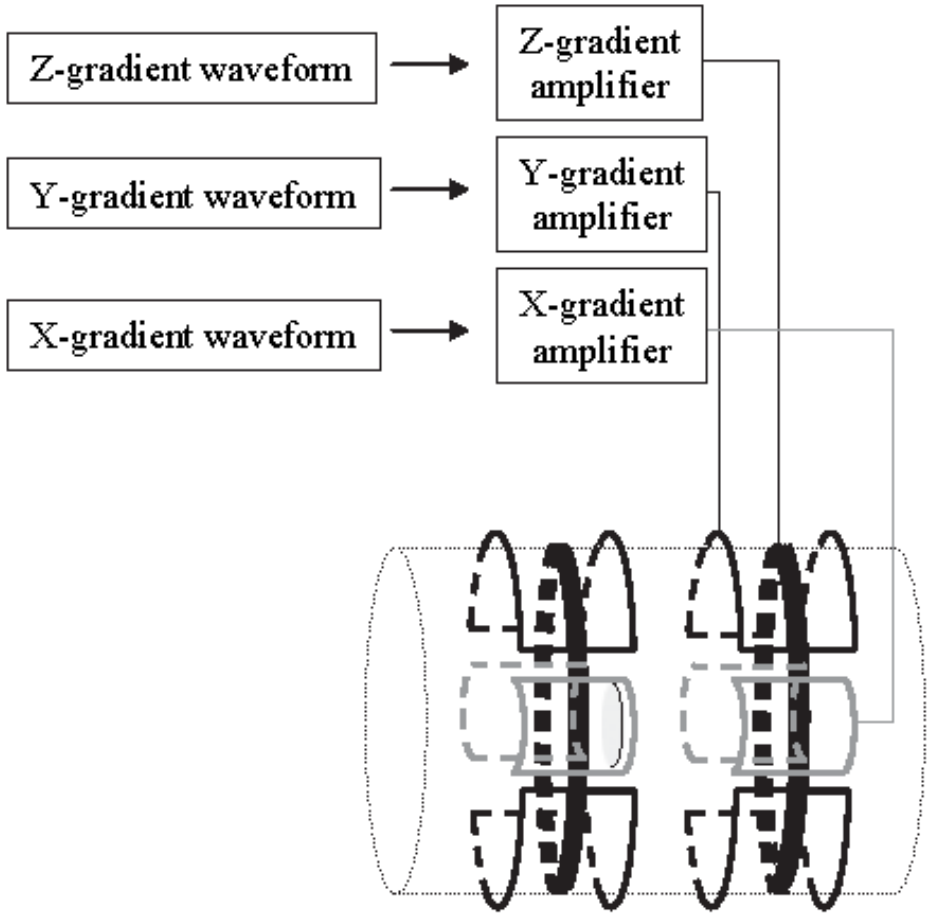
- Produce a uniform variation in magnetic field
  - Depends on coil configuration
- Three sets of coils
  - For *x*, *y*, and *z* directions
  - Precisely wound to control the field
- Important characteristics
  - Slew rate
  - Maximum gradient strength





Gradient Coils cont.

- Typical design

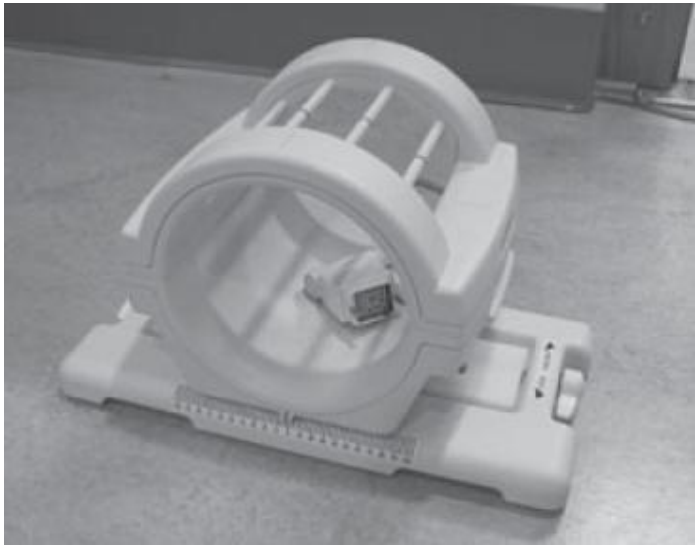


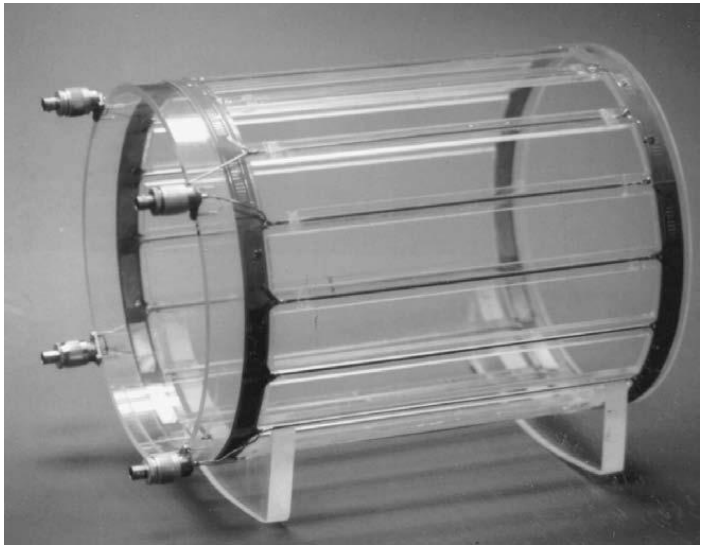
Radio Frequency (RF) Coils

- Act as the antenna
  - Transmit RF signal to the patient and/or
  - Receive the return signal
- Transmit coils
  - Excite sample
  - Uniform excitation of selected region is required
- Receive coils
  - Receive RF signal from relaxing spins

Radio Frequency (RF) Coils

- The same coil can be used for both transmit and receive
  - But often separate coils are used
    - Eg. Body transmit, surface receive

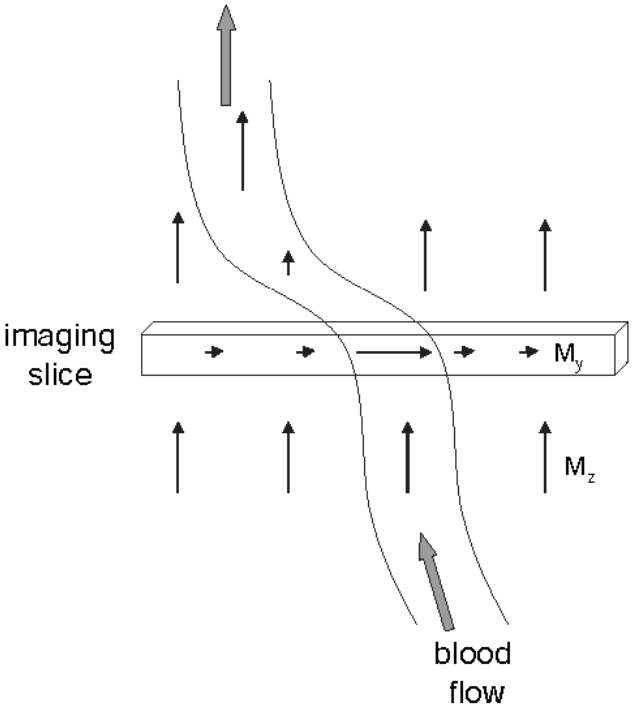


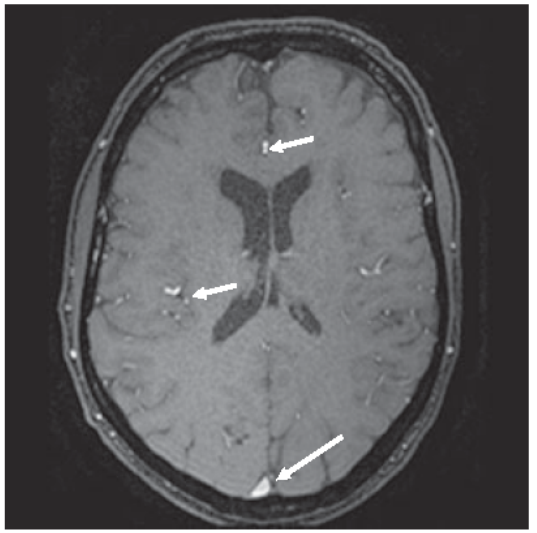


Magnetic Resonance Angiography

- Does not require the use of contrast agents
- Time-of-flight (TOF) angiography
  - The most common technique
  - Based on the much shorter effective T_1 ($T_{1,eff}$) of blood due to flow into and through the imaging slice
  - Uses a rapid gradient-echo sequence with a tip angle large compared to the Ernst angle for tissue

Magnetic Resonance Angiography cont.

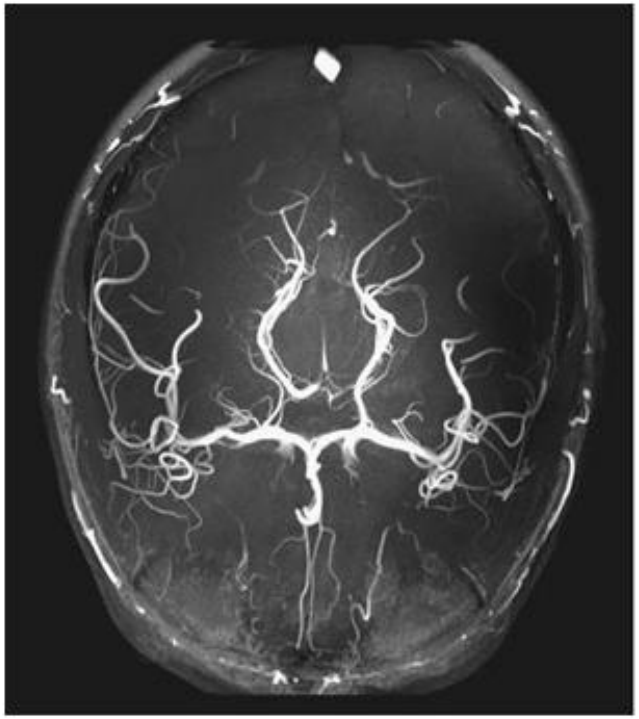


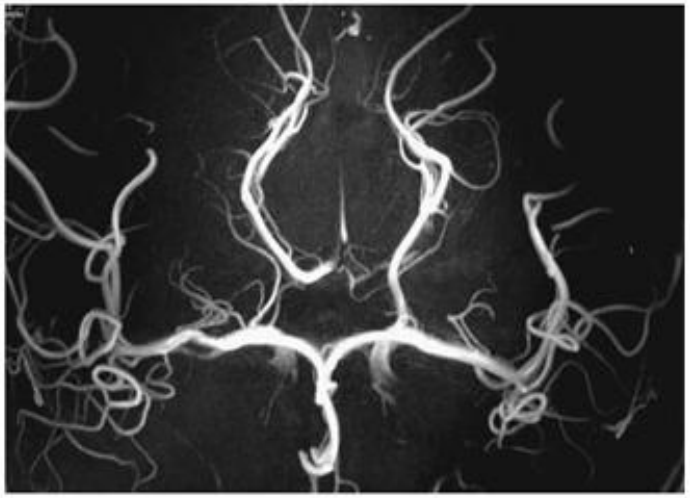


(left) The TOF MR angiographic technique. Blood flowing into the imaging slice has full M_z magnetization before the RF pulse, and so a 90° pulse creates maximum M_y magnetization. In contrast, stationary tissue within the slice experiences every RF pulse during the imaging sequence and since TR $\ll 3T_1$ is 'saturated', i.e. has an M_z magnetization which is much less than its thermal equilibrium value. Therefore, the 90° pulse creates a much lower M_y magnetization in the tissue than in the blood. (right) Image of a slice through the brain showing the vessels (arrows) with flow perpendicular to the image slice as bright spots.

Magnetic Resonance Angiography

- Maximum intensity projection from a three-dimensional TOF sequence acquired in the brain





Functional MRI (fMRI)

- Used to determine which areas of the brain are involved in specific cognitive tasks
- Blood oxygen level dependent (BOLD) effect
  - MRI signal intensity changes depending upon the level of oxygenation of the blood in the brain
- When neuronal activation occurs in the brain's grey matter
  - Local blood flow in the capillary bed is increased
  - The level of oxygenation changes by small amounts which can be detected by MRI
  - Corresponding to a decrease in the concentration of deoxyhaemoglobin

Functional MRI (fMRI) cont.

- Deoxyhaemoglobin
  - Has a magnetic susceptibility which is more paramagnetic than tissue
  - The local magnetic field gradients between the blood in the capillary bed and tissue also decrease
  - Value of T_2 increase locally in areas of the brain which are associated with neuronal activation
  - Increases MR signal on T_2 weighted scans

- The most common MR sequence used is a multi-slice echo-planar imaging
  - Data acquisition is fast enough to obtain whole-brain coverage in a few seconds

Functional MRI (fMRI) cont.

- Data acquisition and processing
  - Signal variations are very small (typically 0.2–2%)
  - Experiments are repeated a number of times with periods of rest between each stimulation block
  - Correlates MRI signal intensity for each pixel with the stimulation waveform followed by statistical analysis to determine whether the correlation is significant

- Currently used more for research than routine clinical use
  - Clinically used in a number of pre-surgical planning studies

Functional MRI (fMRI) cont.

Left: Plot of the MR signal intensity within an activated voxel as a function of time, and stimulus

Right: statistically significant correlation in two areas outlined by the arrows

